

MASTER_CONSOLIDATED_REPORT.txt

TBD2 pipeline output - rendered to PDF

MASTER REPORT: Market and Sales Module Unified Report

Total sections: 8

=== low_performing_entities_report_mkts.txt ===

Low-Performing Metrics

S.No,Metric Id,Metric Name,Metric Value

1,BRM10116,Brand Relevance,44.18333333333333
2,BRM10106,Market Share,44.20933333333333
3,BRM10111,Customer Retention Rate,44.247
4,CHM10103,Channel Cost Efficiency,44.249
5,LYP10114,Cost per Acquisition (CPA),44.31133333333333
6,MKC10118,Time on Page,44.37933333333335
7,SFO10107,Average Deal Size Forecast,44.43866666666667
8,BRM10105,Customer Satisfaction (CSAT),44.501
9,BRM10109,Social Media Engagement,44.538
10,ADS10102,Impressions,44.54433333333334

Low-Performing Groups

S.No,Group Id,Group Name,Group Value

1,CHM10001,Channel Management Metrics,44.8455
2,BRM10001,Brand Management Metrics,44.85613666666666
3,SFO10001,Sales Forecasting Metrics,44.87873333333333
4,MSR10001,Market Sales Support & Readiness Metrics,44.97321666666666
5,MKR10001,Market Research Metrics,44.97335666666667
6,ADS10001,Advertising Metrics,44.98809666666666
7,LYP10001,Loyalty Program Management Metrics,45.01000333333333
8,SDV10001,Sales Development Metrics,45.02077333333334
9,CPG10001,Marketing Campaign Management Metrics,45.02107666666667
10,MKC10001,Marketing Communications Metrics,45.04776

Low-Performing Sub-Modules

S.No,	Sub-Module Id,	Sub-Module Name,	Sub-Module Value
1,	CHM10000,	Channel Management,	44.8455
2,	BRM10000,	Brand Management,	44.856136666666666
3,	SFO10000,	Sales Forecasting,	44.878733333333333
4,	MSR10000,	Market Sales Support & Readiness,	44.973216666666666
5,	MKR10000,	Market Research,	44.973356666666667
6,	ADS10000,	Advertising,	44.988096666666666
7,	LYP10000,	Loyalty Program Management,	45.010003333333333
8,	SDV10000,	Sales Development,	45.020773333333334
9,	CPG10000,	Marketing Campaign Management,	45.021076666666667
10,	MKC10000,	Marketing Communications,	45.04776
Overall Module Average,MKTS_001,			44.98

=== M_G_SM_correlation_report_mkts.txt ===

Correlation Report (top pairs by |correlation|; full matrices are still used internally for CHAID risk binning - see chaid_risk_segmentation_report)

Metrics Correlation Summary

Total pairs examined: 18336
Pairs with |correlation| >= 0.5: 0
Mean |correlation|: 0.0146

No pairs met the threshold - showing top 150 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

BRM10105	Customer Satisfaction (CSAT)	MKR10114	Market Potential	-0.1049
CHM10106	Customer Acquisition by Channel	SDV10117	Sales Forecast Accuracy	0.0748
SSP10103	Conversion Rate	SFO10102	Forecasting Horizon	0.0746
MSP10106	Market Penetration	ADS10105	Cost Per Click (CPC)	-0.0671
BRM10108	Advertising Recall	CPG10119	Email Click-Through Rate	-0.0671
MKR10118	Customer Lifetime Value (CLV)	LYP10116	Loyalty Program Fraud Rate	-0.0664
MKR10117	Customer Churn Rate	ADS10119	Ad Viewability Rate	-0.0647
SDV10114	Upsell and Cross-Sell Rates	MKC10103	Click-Through Rate (CTR)	0.0646
LYP10107	Customer Satisfaction (CSAT)	ADS10108	Ad Engagement Rate	-0.0637
MKC10102	Engagement Rate	ADS10113	Customer Acquisition Cost (CAC)	-0.0629
SFO10105	Sales Pipeline Coverage	MKC10106	Open Rate	-0.0628
SSP10110	Sales Forecast Accuracy	MKC10103	Click-Through Rate (CTR)	-0.0623
SDV10111	Sales Opportunity Volume	MSR10119	Market Feedback Loop Efficiency	0.0622
MKR10113	Price Sensitivity	LYP10112	Referral Rate	-0.0620

CHM10108 | Sales Conversion Rate by Channel | BRM10108 | Advertising Recall | -0.0617
MSR10119 | Market Feedback Loop Efficiency | ADS10106 | Cost Per Thousand Impressions (CPM) | -0.0612
LYP10110 | Churn Rate | LYP10111 | Engagement Rate | -0.0603
SSP10106 | Customer Acquisition Cost (CAC) | CPG10102 | Impressions | -0.0602
CPG10113 | Campaign ROI | CPG10114 | Customer Satisfaction (CSAT) | 0.0601
SDV10109 | Average Revenue per User (ARPU) | MKC10111 | Brand Awareness Increase | -0.0600
SSP10110 | Sales Forecast Accuracy | SDV10102 | Lead Generation Rate | 0.0595
CPG10114 | Customer Satisfaction (CSAT) | MKC10117 | Customer Retention Rate | 0.0591
SFO10107 | Average Deal Size Forecast | BRM10116 | Brand Relevance | -0.0589
MSP10105 | Sales Growth | CHM10101 | Channel Revenue | 0.0588
LYP10116 | Loyalty Program Fraud Rate | MKC10103 | Click-Through Rate (CTR) | -0.0586
SFO10104 | Lead Time | CPG10111 | Engagement Rate | 0.0584
SSP10105 | Sales Cycle Length | MKC10103 | Click-Through Rate (CTR) | 0.0571
SDV10109 | Average Revenue per User (ARPU) | CPG10118 | Email Open Rate | 0.0568
MKR10119 | Return on Market Research Investment (ROMRI) | SDV10102 | Lead Generation Rate | 0.0564
CPG10105 | Lead Generation Rate | CPG10114 | Customer Satisfaction (CSAT) | 0.0562
MSP10105 | Sales Growth | MKR10119 | Return on Market Research Investment (ROMRI) | -0.0555
BRM10110 | Brand Sentiment | ADS10107 | Return on Advertising Spend (ROAS) | 0.0554
LYP10104 | Customer Lifetime Value (CLV) | LYP10112 | Referral Rate | -0.0551
MKR10114 | Market Potential | CPG10112 | Bounce Rate | -0.0550
CHM10110 | Channel Satisfaction | LYP10106 | Net Promoter Score (NPS) | 0.0549
SSP10106 | Customer Acquisition Cost (CAC) | MSR10118 | Sales Support Team Productivity | -0.0549
LYP10105 | Repeat Purchase Rate | MKC10106 | Open Rate | 0.0549
SDV10102 | Lead Generation Rate | ADS10101 | Reach | -0.0546
SDV10111 | Sales Opportunity Volume | LYP10113 | Customer Feedback Score | 0.0546
SDV10114 | Upsell and Cross-Sell Rates | LYP10116 | Loyalty Program Fraud Rate | -0.0542
MKR10113 | Price Sensitivity | MKC10102 | Engagement Rate | -0.0541
SDV10103 | Lead Conversion Rate | LYP10111 | Engagement Rate | -0.0541
MSP10108 | Campaign Effectiveness | CHM10108 | Sales Conversion Rate by Channel | -0.0538
MKR10107 | Net Promoter Score (NPS) | MKC10109 | Share of Voice | -0.0538
CPG10117 | Social Media Mentions | MKC10117 | Customer Retention Rate | -0.0537
SSP10102 | Sales Pipeline Value | ADS10107 | Return on Advertising Spend (ROAS) | 0.0537
SDV10102 | Lead Generation Rate | MKC10112 | Lead Generation Rate | -0.0536
SSP10101 | Sales Target Achievement | SFO10107 | Average Deal Size Forecast | 0.0536
MKC10109 | Share of Voice | ADS10102 | Impressions | -0.0535
MKR10109 | Product Performance | MSR10116 | Knowledge Base Usage | 0.0534
MSP10110 | Customer Satisfaction | CHM10106 | Customer Acquisition by Channel | 0.0533
CPG10111 | Engagement Rate | CPG10113 | Campaign ROI | -0.0533
SFO10103 | Sales Forecast Variance | MSR10108 | Sales Support Ticket Resolution Time | -0.0533
LYP10116 | Loyalty Program Fraud Rate | CPG10108 | Sales Revenue | -0.0532

SFO10101 | Forecast Accuracy | SDV10119 | Revenue Growth from New Products | 0.0531
CHM10106 | Customer Acquisition by Channel | CPG10117 | Social Media Mentions | -0.0530
SFO10109 | Seasonal Adjustments | MSR10107 | Customer Support Response Time | 0.0528
LYP10103 | Reward Redemption Rate | CPG10116 | Brand Awareness Increase | 0.0527
MKR10102 | Market Share by Segment | MKC10113 | Cost Per Click (CPC) | 0.0525
MSP10101 | Market Share | BRM10105 | Customer Satisfaction (CSAT) | -0.0523
SDV10115 | Customer Retention Rate | ADS10118 | Video View Rate | 0.0523
MKC10119 | Content Share Rate | ADS10110 | Bounce Rate | 0.0522
MSP10104 | Customer Lifetime Value (CLV) | MKC10112 | Lead Generation Rate | 0.0521
SSP10103 | Conversion Rate | BRM10101 | Brand Awareness | 0.0521
MSP10101 | Market Share | ADS10106 | Cost Per Thousand Impressions (CPM) | -0.0519
SSP10110 | Sales Forecast Accuracy | LYP10116 | Loyalty Program Fraud Rate | 0.0518
MKR10101 | Market Size and Growth | CPG10113 | Campaign ROI | -0.0517
SSP10104 | Average Deal Size | CHM10102 | Channel Profitability | -0.0517
SSP10107 | Customer Retention Rate | CPG10110 | Customer Retention Rate | 0.0517
BRM10102 | Brand Perception | ADS10105 | Cost Per Click (CPC) | -0.0515
SDV10119 | Revenue Growth from New Products | LYP10117 | Loyalty Program Tier Upgrades | -0.0515
SSP10106 | Customer Acquisition Cost (CAC) | ADS10114 | Lead Generation Rate | -0.0515
SDV10107 | Sales Team Productivity | CPG10111 | Engagement Rate | -0.0515
CHM10110 | Channel Satisfaction | ADS10101 | Reach | -0.0515
MSP10102 | Customer Acquisition Cost (CAC) | MSR10102 | Training Completion Rate | -0.0515
CPG10117 | Social Media Mentions | MKC10106 | Open Rate | 0.0514
SDV10116 | Churn Rate | LYP10119 | Loyalty Points Expiration Rate | 0.0514
CHM10106 | Customer Acquisition by Channel | CPG10116 | Brand Awareness Increase | 0.0514
SSP10105 | Sales Cycle Length | CHM10110 | Channel Satisfaction | -0.0512
SSP10107 | Customer Retention Rate | LYP10110 | Churn Rate | -0.0512
MSR10118 | Sales Support Team Productivity | CPG10109 | Customer Acquisition Cost (CAC) | 0.0512
MSR10109 | Sales Support Satisfaction | ADS10112 | Ad Recall | -0.0511
CHM10104 | Channel Penetration | CPG10112 | Bounce Rate | 0.0511
SDV10112 | New Account Acquisition | MSR10106 | Lead Response Time | -0.0511
SSP10104 | Average Deal Size | SDV10107 | Sales Team Productivity | -0.0510
MKR10103 | Customer Demographics | CPG10105 | Lead Generation Rate | 0.0508
MSP10102 | Customer Acquisition Cost (CAC) | CHM10107 | Customer Retention by Channel | 0.0508
SSP10105 | Sales Cycle Length | SDV10116 | Churn Rate | 0.0508
BRM10117 | Customer Acquisition Cost (CAC) | CPG10110 | Customer Retention Rate | 0.0507
MSP10107 | Brand Awareness | SSP10110 | Sales Forecast Accuracy | 0.0507
BRM10115 | Brand Consistency | MSR10119 | Market Feedback Loop Efficiency | 0.0507
SDV10113 | Sales Training Effectiveness | SDV10117 | Sales Forecast Accuracy | 0.0507
MKC10114 | Customer Feedback Score | MKC10116 | Return on Marketing Investment (ROMI) | 0.0506
SSP10105 | Sales Cycle Length | BRM10109 | Social Media Engagement | -0.0506

BRM10105 | Customer Satisfaction (CSAT) | SDV10109 | Average Revenue per User (ARPU) | 0.0505
BRM10102 | Brand Perception | MSR10118 | Sales Support Team Productivity | -0.0505
MKR10113 | Price Sensitivity | MSR10112 | Content Effectiveness | 0.0503
CPG10118 | Email Open Rate | ADS10101 | Reach | 0.0503
MSR10105 | Sales Tool Adoption Rate | ADS10115 | Ad Sentiment Analysis | -0.0503
MKC10114 | Customer Feedback Score | ADS10109 | Ad Frequency | 0.0503
SDV10112 | New Account Acquisition | MSR10117 | Sales Collateral Production Time | -0.0502
LYP10112 | Referral Rate | MKC10106 | Open Rate | -0.0500
SFO10104 | Lead Time | MSR10106 | Lead Response Time | -0.0500
SDV10106 | Time to Close (Sales Cycle Length) | CPG10115 | Net Promoter Score (NPS) | 0.0499
BRM10116 | Brand Relevance | ADS10106 | Cost Per Thousand Impressions (CPM) | 0.0499
LYP10118 | Inactive Member Rate | ADS10105 | Cost Per Click (CPC) | 0.0498
BRM10101 | Brand Awareness | MKC10118 | Time on Page | -0.0498
MSP10106 | Market Penetration | LYP10113 | Customer Feedback Score | 0.0498
MSR10116 | Knowledge Base Usage | LYP10103 | Reward Redemption Rate | -0.0497
CHM10107 | Customer Retention by Channel | MSR10117 | Sales Collateral Production Time | 0.0497
SFO10107 | Average Deal Size Forecast | LYP10105 | Repeat Purchase Rate | 0.0497
CHM10103 | Channel Cost Efficiency | BRM10109 | Social Media Engagement | -0.0497
LYP10104 | Customer Lifetime Value (CLV) | ADS10103 | Click-Through Rate (CTR) | -0.0496
BRM10107 | Brand Loyalty | MSR10119 | Market Feedback Loop Efficiency | -0.0495
SSP10107 | Customer Retention Rate | ADS10116 | Ad Share of Voice | 0.0494
SDV10102 | Lead Generation Rate | SDV10105 | Customer Acquisition Cost (CAC) | 0.0493
CPG10114 | Customer Satisfaction (CSAT) | MKC10101 | Message Reach | -0.0493
MKR10102 | Market Share by Segment | ADS10118 | Video View Rate | -0.0493
SDV10119 | Revenue Growth from New Products | MSR10117 | Sales Collateral Production Time | -0.0492
MSR10105 | Sales Tool Adoption Rate | MKC10109 | Share of Voice | -0.0492
CHM10110 | Channel Satisfaction | MSR10117 | Sales Collateral Production Time | -0.0492
SFO10101 | Forecast Accuracy | MKR10119 | Return on Market Research Investment (ROMRI) | 0.0491
MKC10119 | Content Share Rate | ADS10116 | Ad Share of Voice | 0.0491
CHM10102 | Channel Profitability | CHM10103 | Channel Cost Efficiency | 0.0491
LYP10102 | Loyalty Program Enrollment Rate | CPG10108 | Sales Revenue | -0.0490
MSR10106 | Lead Response Time | MSR10111 | Market Readiness Score | -0.0489
BRM10107 | Brand Loyalty | BRM10112 | Brand Advocacy | 0.0489
BRM10111 | Customer Retention Rate | SDV10118 | Market Penetration Rate | -0.0489
BRM10111 | Customer Retention Rate | MSR10102 | Training Completion Rate | 0.0488
SSP10103 | Conversion Rate | BRM10112 | Brand Advocacy | -0.0488
MSR10112 | Content Effectiveness | ADS10115 | Ad Sentiment Analysis | -0.0488
MSP10107 | Brand Awareness | MKC10106 | Open Rate | 0.0484
MSP10110 | Customer Satisfaction | SDV10107 | Sales Team Productivity | 0.0484
MKR10117 | Customer Churn Rate | MKC10110 | Sentiment Analysis | 0.0484

MSP10102 | Customer Acquisition Cost (CAC) | MKR10118 | Customer Lifetime Value (CLV) | 0.0484
SSP10105 | Sales Cycle Length | MKR10111 | Consumer Awareness and Perception | 0.0484
MKR10107 | Net Promoter Score (NPS) | MKC10112 | Lead Generation Rate | -0.0484
ADS10111 | Conversion Value | ADS10119 | Ad Viewability Rate | 0.0483
BRM10105 | Customer Satisfaction (CSAT) | MKR10111 | Consumer Awareness and Perception | -0.0483
MSR10117 | Sales Collateral Production Time | ADS10117 | Social Media Mentions | -0.0483
MSP10108 | Campaign Effectiveness | SFO10108 | Sales Growth Forecast | 0.0482
BRM10117 | Customer Acquisition Cost (CAC) | MSR10110 | Sales Support Ticket Volume | -0.0481
BRM10107 | Brand Loyalty | MKR10101 | Market Size and Growth | -0.0481
MSR10116 | Knowledge Base Usage | LYP10102 | Loyalty Program Enrollment Rate | 0.0481
LYP10119 | Loyalty Points Expiration Rate | MKC10116 | Return on Marketing Investment (ROMI) | 0.0481
SDV10106 | Time to Close (Sales Cycle Length) | MKC10113 | Cost Per Click (CPC) | -0.0481
BRM10114 | Brand Preference | BRM10117 | Customer Acquisition Cost (CAC) | 0.0480
SFO10109 | Seasonal Adjustments | ADS10112 | Ad Recall | 0.0480
BRM10116 | Brand Relevance | ADS10105 | Cost Per Click (CPC) | -0.0480
BRM10112 | Brand Advocacy | SDV10106 | Time to Close (Sales Cycle Length) | -0.0480

Groups Correlation Summary

Total pairs examined: 66

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0163

No pairs met the threshold - showing top 66 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

SSP10001 | Sales Strategy and Planning Metrics | CHM10001 | Channel Management Metrics | -0.0541
MKR10001 | Market Research Metrics | ADS10001 | Advertising Metrics | -0.0524
MSP10001 | Market Strategy and Policy Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0397
MSP10001 | Market Strategy and Policy Metrics | SDV10001 | Sales Development Metrics | 0.0366
SDV10001 | Sales Development Metrics | ADS10001 | Advertising Metrics | 0.0358
SFO10001 | Sales Forecasting Metrics | SDV10001 | Sales Development Metrics | 0.0328
MKR10001 | Market Research Metrics | SDV10001 | Sales Development Metrics | -0.0321
CPG10001 | Marketing Campaign Management Metrics | MKC10001 | Marketing Communications Metrics | 0.0307
CHM10001 | Channel Management Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0307
SDV10001 | Sales Development Metrics | MKC10001 | Marketing Communications Metrics | 0.0294
MKC10001 | Marketing Communications Metrics | ADS10001 | Advertising Metrics | -0.0285
LYP10001 | Loyalty Program Management Metrics | ADS10001 | Advertising Metrics | -0.0280
MSR10001 | Market Sales Support & Readiness Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0272
SSP10001 | Sales Strategy and Planning Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0267
SSP10001 | Sales Strategy and Planning Metrics | ADS10001 | Advertising Metrics | 0.0267

SFO10001 | Sales Forecasting Metrics | LYP10001 | Loyalty Program Management Metrics | 0.0267
SSP10001 | Sales Strategy and Planning Metrics | MKR10001 | Market Research Metrics | -0.0263
BRM10001 | Brand Management Metrics | MKC10001 | Marketing Communications Metrics | -0.0241
BRM10001 | Brand Management Metrics | MKR10001 | Market Research Metrics | 0.0233
SSP10001 | Sales Strategy and Planning Metrics | BRM10001 | Brand Management Metrics | -0.0212
SSP10001 | Sales Strategy and Planning Metrics | MSR10001 | Market Sales Support & Readiness Metrics | 0.0207
MSP10001 | Market Strategy and Policy Metrics | BRM10001 | Brand Management Metrics | -0.0206
SFO10001 | Sales Forecasting Metrics | CPG10001 | Marketing Campaign Management Metrics | 0.0205
MSP10001 | Market Strategy and Policy Metrics | SSP10001 | Sales Strategy and Planning Metrics | 0.0200
MKR10001 | Market Research Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0186
SDV10001 | Sales Development Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0183
LYP10001 | Loyalty Program Management Metrics | CPG10001 | Marketing Campaign Management Metrics | 0.0180
SFO10001 | Sales Forecasting Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0177
MSP10001 | Market Strategy and Policy Metrics | CHM10001 | Channel Management Metrics | -0.0174
SDV10001 | Sales Development Metrics | LYP10001 | Loyalty Program Management Metrics | -0.0172
SSP10001 | Sales Strategy and Planning Metrics | SDV10001 | Sales Development Metrics | 0.0166
BRM10001 | Brand Management Metrics | SDV10001 | Sales Development Metrics | -0.0157
SDV10001 | Sales Development Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0144
CHM10001 | Channel Management Metrics | MKC10001 | Marketing Communications Metrics | 0.0142
CHM10001 | Channel Management Metrics | BRM10001 | Brand Management Metrics | -0.0136
BRM10001 | Brand Management Metrics | LYP10001 | Loyalty Program Management Metrics | -0.0129
MKR10001 | Market Research Metrics | LYP10001 | Loyalty Program Management Metrics | 0.0115
CHM10001 | Channel Management Metrics | MKR10001 | Market Research Metrics | 0.0113
SFO10001 | Sales Forecasting Metrics | MKC10001 | Marketing Communications Metrics | 0.0113
CHM10001 | Channel Management Metrics | SFO10001 | Sales Forecasting Metrics | 0.0105
LYP10001 | Loyalty Program Management Metrics | MKC10001 | Marketing Communications Metrics | 0.0091
SSP10001 | Sales Strategy and Planning Metrics | LYP10001 | Loyalty Program Management Metrics | 0.0088
BRM10001 | Brand Management Metrics | CPG10001 | Marketing Campaign Management Metrics | 0.0086
MSP10001 | Market Strategy and Policy Metrics | MKC10001 | Marketing Communications Metrics | 0.0085
SFO10001 | Sales Forecasting Metrics | MKR10001 | Market Research Metrics | 0.0085
CHM10001 | Channel Management Metrics | ADS10001 | Advertising Metrics | -0.0074
MSP10001 | Market Strategy and Policy Metrics | MKR10001 | Market Research Metrics | 0.0072
CHM10001 | Channel Management Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0065
MSR10001 | Market Sales Support & Readiness Metrics | MKC10001 | Marketing Communications Metrics | 0.0061
BRM10001 | Brand Management Metrics | ADS10001 | Advertising Metrics | 0.0061
MSP10001 | Market Strategy and Policy Metrics | CPG10001 | Marketing Campaign Management Metrics | -0.0054
SSP10001 | Sales Strategy and Planning Metrics | MKC10001 | Marketing Communications Metrics | -0.0047
CPG10001 | Marketing Campaign Management Metrics | ADS10001 | Advertising Metrics | 0.0043

CHM10001 | Channel Management Metrics | LYP10001 | Loyalty Program Management Metrics | -0.0042
 MSR10001 | Market Sales Support & Readiness Metrics | ADS10001 | Advertising Metrics | 0.0038
 SSP10001 | Sales Strategy and Planning Metrics | SFO10001 | Sales Forecasting Metrics | 0.0029
 SFO10001 | Sales Forecasting Metrics | BRM10001 | Brand Management Metrics | -0.0028
 SFO10001 | Sales Forecasting Metrics | ADS10001 | Advertising Metrics | -0.0022
 MSP10001 | Market Strategy and Policy Metrics | SFO10001 | Sales Forecasting Metrics | -0.0022
 MSR10001 | Market Sales Support & Readiness Metrics | LYP10001 | Loyalty Program Management Metrics | -0.0016
 MSP10001 | Market Strategy and Policy Metrics | ADS10001 | Advertising Metrics | -0.0015
 CHM10001 | Channel Management Metrics | SDV10001 | Sales Development Metrics | 0.0015
 MKR10001 | Market Research Metrics | MKC10001 | Marketing Communications Metrics | -0.0015
 MKR10001 | Market Research Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0012
 BRM10001 | Brand Management Metrics | MSR10001 | Market Sales Support & Readiness Metrics | -0.0011
 MSP10001 | Market Strategy and Policy Metrics | LYP10001 | Loyalty Program Management Metrics | 0.0008

Sub_Modules Correlation Summary

Total pairs examined: 66

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0163

No pairs met the threshold - showing top 66 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

SSP10000 | Sales Strategy and Planning | CHM10000 | Channel Management | -0.0541
 MKR10000 | Market Research | ADS10000 | Advertising | -0.0524
 MSP10000 | Market Strategy and Policy | MSR10000 | Market Sales Support & Readiness | -0.0397
 MSP10000 | Market Strategy and Policy | SDV10000 | Sales Development | 0.0366
 SDV10000 | Sales Development | ADS10000 | Advertising | 0.0358
 SFO10000 | Sales Forecasting | SDV10000 | Sales Development | 0.0328
 MKR10000 | Market Research | SDV10000 | Sales Development | -0.0321
 CPG10000 | Marketing Campaign Management | MKC10000 | Marketing Communications | 0.0307
 CHM10000 | Channel Management | MSR10000 | Market Sales Support & Readiness | -0.0307
 SDV10000 | Sales Development | MKC10000 | Marketing Communications | 0.0294
 MKC10000 | Marketing Communications | ADS10000 | Advertising | -0.0285
 LYP10000 | Loyalty Program Management | ADS10000 | Advertising | -0.0280
 MSR10000 | Market Sales Support & Readiness | CPG10000 | Marketing Campaign Management | -0.0272
 SSP10000 | Sales Strategy and Planning | CPG10000 | Marketing Campaign Management | -0.0267
 SSP10000 | Sales Strategy and Planning | ADS10000 | Advertising | 0.0267
 SFO10000 | Sales Forecasting | LYP10000 | Loyalty Program Management | 0.0267
 SSP10000 | Sales Strategy and Planning | MKR10000 | Market Research | -0.0263
 BRM10000 | Brand Management | MKC10000 | Marketing Communications | -0.0241
 BRM10000 | Brand Management | MKR10000 | Market Research | 0.0233

SSP10000 | Sales Strategy and Planning | BRM10000 | Brand Management | -0.0212
SSP10000 | Sales Strategy and Planning | MSR10000 | Market Sales Support & Readiness | 0.0207
MSP10000 | Market Strategy and Policy | BRM10000 | Brand Management | -0.0206
SFO10000 | Sales Forecasting | CPG10000 | Marketing Campaign Management | 0.0205
MSP10000 | Market Strategy and Policy | SSP10000 | Sales Strategy and Planning | 0.0200
MKR10000 | Market Research | CPG10000 | Marketing Campaign Management | -0.0186
SDV10000 | Sales Development | CPG10000 | Marketing Campaign Management | -0.0183
LYP10000 | Loyalty Program Management | CPG10000 | Marketing Campaign Management | 0.0180
SFO10000 | Sales Forecasting | MSR10000 | Market Sales Support & Readiness | -0.0177
MSP10000 | Market Strategy and Policy | CHM10000 | Channel Management | -0.0174
SDV10000 | Sales Development | LYP10000 | Loyalty Program Management | -0.0172
SSP10000 | Sales Strategy and Planning | SDV10000 | Sales Development | 0.0166
BRM10000 | Brand Management | SDV10000 | Sales Development | -0.0157
SDV10000 | Sales Development | MSR10000 | Market Sales Support & Readiness | -0.0144
CHM10000 | Channel Management | MKC10000 | Marketing Communications | 0.0142
CHM10000 | Channel Management | BRM10000 | Brand Management | -0.0136
BRM10000 | Brand Management | LYP10000 | Loyalty Program Management | -0.0129
MKR10000 | Market Research | LYP10000 | Loyalty Program Management | 0.0115
CHM10000 | Channel Management | MKR10000 | Market Research | 0.0113
SFO10000 | Sales Forecasting | MKC10000 | Marketing Communications | 0.0113
CHM10000 | Channel Management | SFO10000 | Sales Forecasting | 0.0105
LYP10000 | Loyalty Program Management | MKC10000 | Marketing Communications | 0.0091
SSP10000 | Sales Strategy and Planning | LYP10000 | Loyalty Program Management | 0.0088
BRM10000 | Brand Management | CPG10000 | Marketing Campaign Management | 0.0086
MSP10000 | Market Strategy and Policy | MKC10000 | Marketing Communications | 0.0085
SFO10000 | Sales Forecasting | MKR10000 | Market Research | 0.0085
CHM10000 | Channel Management | ADS10000 | Advertising | -0.0074
MSP10000 | Market Strategy and Policy | MKR10000 | Market Research | 0.0072
CHM10000 | Channel Management | CPG10000 | Marketing Campaign Management | -0.0065
MSR10000 | Market Sales Support & Readiness | MKC10000 | Marketing Communications | 0.0061
BRM10000 | Brand Management | ADS10000 | Advertising | 0.0061
MSP10000 | Market Strategy and Policy | CPG10000 | Marketing Campaign Management | -0.0054
SSP10000 | Sales Strategy and Planning | MKC10000 | Marketing Communications | -0.0047
CPG10000 | Marketing Campaign Management | ADS10000 | Advertising | 0.0043
CHM10000 | Channel Management | LYP10000 | Loyalty Program Management | -0.0042
MSR10000 | Market Sales Support & Readiness | ADS10000 | Advertising | 0.0038
SSP10000 | Sales Strategy and Planning | SFO10000 | Sales Forecasting | 0.0029
SFO10000 | Sales Forecasting | BRM10000 | Brand Management | -0.0028
SFO10000 | Sales Forecasting | ADS10000 | Advertising | -0.0022
MSP10000 | Market Strategy and Policy | SFO10000 | Sales Forecasting | -0.0022

MSR10000 | Market Sales Support & Readiness | LYP10000 | Loyalty Program Management | -0.0016
MSP10000 | Market Strategy and Policy | ADS10000 | Advertising | -0.0015
CHM10000 | Channel Management | SDV10000 | Sales Development | 0.0015
MKR10000 | Market Research | MKC10000 | Marketing Communications | -0.0015
MKR10000 | Market Research | MSR10000 | Market Sales Support & Readiness | -0.0012
BRM10000 | Brand Management | MSR10000 | Market Sales Support & Readiness | -0.0011
MSP10000 | Market Strategy and Policy | LYP10000 | Loyalty Program Management | 0.0008

=== chaid_risk_segmentation_report_mkts.txt ===

CHAID Risk Segmentation Report
Generated on: 2026-09-20 15:00:32

CHAID Tree Summary:
[[([], {'Critical': 79.0, 'High': 113.0}, (Repetition, p=1.163552427676054e-43, score=191.99999999999997, groups=[['Irregular'], ['Repetitive']]), dof=1)), ([['Irregular'], {'Critical': 79.0, 'High': 0}], <Invalid Chaid Split> - the node only contains single category respondents), ([['Repetitive'], {'Critical': 0, 'High': 113.0}], <Invalid Chaid Split> - the node only contains single category respondents)]

High-Risk Metrics Summary (Critical + High):

Metric ID | Metric Name | Risk Level | Trend | Repetition | Avg Correlation | Inconsistency Level

MSP10101 | Market Share | Critical | Decreasing | Irregular | Low | Very High
MSP10102 | Customer Acquisition Cost (CAC) | High | Decreasing | Repetitive | Low | Very High
MSP10103 | Customer Retention Rate | High | Stable | Repetitive | Low | Very High
MSP10104 | Customer Lifetime Value (CLV) | High | Increasing | Repetitive | Low | Very High
MSP10105 | Sales Growth | High | Decreasing | Repetitive | Low | Very High
MSP10106 | Market Penetration | High | Decreasing | Repetitive | Low | Very High
MSP10107 | Brand Awareness | High | Decreasing | Repetitive | Low | Very High
MSP10108 | Campaign Effectiveness | Critical | Decreasing | Irregular | Low | Very High
MSP10109 | Return on Marketing Investment (ROMI) | High | Decreasing | Repetitive | Low | Very High
MSP10110 | Customer Satisfaction | High | Decreasing | Repetitive | Low | Very High
SSP10101 | Sales Target Achievement | High | Stable | Repetitive | Low | Very High
SSP10102 | Sales Pipeline Value | Critical | Stable | Irregular | Low | Very High
SSP10103 | Conversion Rate | Critical | Stable | Irregular | Low | Very High
SSP10104 | Average Deal Size | High | Stable | Repetitive | Low | Very High
SSP10105 | Sales Cycle Length | Critical | Stable | Irregular | Low | Very High
SSP10106 | Customer Acquisition Cost (CAC) | High | Increasing | Repetitive | Low | Very High
SSP10107 | Customer Retention Rate | Critical | Decreasing | Irregular | Low | Very High
SSP10108 | Sales Growth | High | Increasing | Repetitive | Low | Very High
SSP10109 | Market Penetration | High | Increasing | Repetitive | Low | Very High
SSP10110 | Sales Forecast Accuracy | High | Decreasing | Repetitive | Low | Very High
CHM10101 | Channel Revenue | High | Stable | Repetitive | Low | Very High

CHM10102 | Channel Profitability | High | Decreasing | Repetitive | Low | Very High
CHM10103 | Channel Cost Efficiency | High | Stable | Repetitive | Low | Very High
CHM10104 | Channel Penetration | High | Decreasing | Repetitive | Low | Very High
CHM10105 | Channel Growth Rate | High | Decreasing | Repetitive | Low | Very High
CHM10106 | Customer Acquisition by Channel | High | Decreasing | Repetitive | Low | Very High
CHM10107 | Customer Retention by Channel | Critical | Decreasing | Irregular | Low | Very High
CHM10108 | Sales Conversion Rate by Channel | Critical | Decreasing | Irregular | Low | Very High
CHM10109 | Average Deal Size by Channel | High | Stable | Repetitive | Low | Very High
CHM10110 | Channel Satisfaction | Critical | Stable | Irregular | Low | Very High
SFO10101 | Forecast Accuracy | High | Stable | Repetitive | Low | Very High
SFO10102 | Forecasting Horizon | Critical | Increasing | Irregular | Low | Very High
SFO10103 | Sales Forecast Variance | High | Decreasing | Repetitive | Low | Very High
SFO10104 | Lead Time | Critical | Decreasing | Irregular | Low | Very High
SFO10105 | Sales Pipeline Coverage | Critical | Decreasing | Irregular | Low | Very High
SFO10106 | Conversion Rate by Stage | Critical | Stable | Irregular | Low | Very High
SFO10107 | Average Deal Size Forecast | High | Stable | Repetitive | Low | Very High
SFO10108 | Sales Growth Forecast | High | Decreasing | Repetitive | Low | Very High
SFO10109 | Seasonal Adjustments | High | Stable | Repetitive | Low | Very High
SFO10110 | Sales Forecasting Methodology | High | Increasing | Repetitive | Low | Very High
BRM10101 | Brand Awareness | Critical | Stable | Irregular | Low | Very High
BRM10102 | Brand Perception | Critical | Decreasing | Irregular | Low | Very High
BRM10103 | Brand Equity | Critical | Increasing | Irregular | Low | Very High
BRM10104 | Net Promoter Score (NPS) | High | Increasing | Repetitive | Low | Very High
BRM10105 | Customer Satisfaction (CSAT) | High | Stable | Repetitive | Low | Very High
BRM10106 | Market Share | High | Decreasing | Repetitive | Low | Very High
BRM10107 | Brand Loyalty | High | Increasing | Repetitive | Low | Very High
BRM10108 | Advertising Recall | High | Increasing | Repetitive | Low | Very High
BRM10109 | Social Media Engagement | High | Stable | Repetitive | Low | Very High
BRM10110 | Brand Sentiment | High | Decreasing | Repetitive | Low | Very High
BRM10111 | Customer Retention Rate | High | Increasing | Repetitive | Low | Very High
BRM10112 | Brand Advocacy | High | Stable | Repetitive | Low | Very High
BRM10113 | Brand Differentiation | High | Increasing | Repetitive | Low | Very High
BRM10114 | Brand Preference | High | Stable | Repetitive | Low | Very High
BRM10115 | Brand Consistency | High | Decreasing | Repetitive | Low | Very High
BRM10116 | Brand Relevance | Critical | Decreasing | Irregular | Low | Very High
BRM10117 | Customer Acquisition Cost (CAC) | Critical | Stable | Irregular | Low | Very High
BRM10118 | Return on Brand Investment (ROBI) | High | Increasing | Repetitive | Low | Very High
BRM10119 | Brand Extension Success | Critical | Increasing | Irregular | Low | Very High
MKR10101 | Market Size and Growth | High | Increasing | Repetitive | Low | Very High
MKR10102 | Market Share by Segment | Critical | Stable | Irregular | Low | Very High

MKR10103 | Customer Demographics | High | Decreasing | Repetitive | Low | Very High
MKR10104 | Customer Preferences and Behavior | High | Stable | Repetitive | Low | Very High
MKR10105 | Competitive Analysis | High | Decreasing | Repetitive | Low | Very High
MKR10106 | Customer Satisfaction (CSAT) | Critical | Stable | Irregular | Low | Very High
MKR10107 | Net Promoter Score (NPS) | Critical | Stable | Irregular | Low | Very High
MKR10108 | Brand Perception | Critical | Increasing | Irregular | Low | Very High
MKR10109 | Product Performance | High | Increasing | Repetitive | Low | Very High
MKR10110 | Market Trends and Insights | High | Stable | Repetitive | Low | Very High
MKR10111 | Consumer Awareness and Perception | High | Decreasing | Repetitive | Low | Very High
MKR10112 | Usage and Attitude Studies | High | Stable | Repetitive | Low | Very High
MKR10113 | Price Sensitivity | Critical | Stable | Irregular | Low | Very High
MKR10114 | Market Potential | Critical | Increasing | Irregular | Low | Very High
MKR10115 | Regulatory Impact | High | Decreasing | Repetitive | Low | Very High
MKR10116 | Channel Effectiveness | Critical | Increasing | Irregular | Low | Very High
MKR10117 | Customer Churn Rate | Critical | Stable | Irregular | Low | Very High
MKR10118 | Customer Lifetime Value (CLV) | Critical | Stable | Irregular | Low | Very High
MKR10119 | Return on Market Research Investment (ROMRI) | Critical | Decreasing | Irregular | Low | Very High
SDV10101 | Sales Growth Rate | Critical | Stable | Irregular | Low | Very High
SDV10102 | Lead Generation Rate | Critical | Decreasing | Irregular | Low | Very High
SDV10103 | Lead Conversion Rate | High | Stable | Repetitive | Low | Very High
SDV10104 | Sales Qualified Leads (SQLs) | Critical | Decreasing | Irregular | Low | Very High
SDV10105 | Customer Acquisition Cost (CAC) | Critical | Increasing | Irregular | Low | Very High
SDV10106 | Time to Close (Sales Cycle Length) | Critical | Stable | Irregular | Low | Very High
SDV10107 | Sales Team Productivity | High | Decreasing | Repetitive | Low | Very High
SDV10108 | Pipeline Velocity | Critical | Increasing | Irregular | Low | Very High
SDV10109 | Average Revenue per User (ARPU) | Critical | Stable | Irregular | Low | Very High
SDV10110 | Win Rate | High | Stable | Repetitive | Low | Very High
SDV10111 | Sales Opportunity Volume | Critical | Decreasing | Irregular | Low | Very High
SDV10112 | New Account Acquisition | High | Decreasing | Repetitive | Low | Very High
SDV10113 | Sales Training Effectiveness | Critical | Stable | Irregular | Low | Very High
SDV10114 | Upsell and Cross-Sell Rates | High | Decreasing | Repetitive | Low | Very High
SDV10115 | Customer Retention Rate | Critical | Decreasing | Irregular | Low | Very High
SDV10116 | Churn Rate | Critical | Decreasing | Irregular | Low | Very High
SDV10117 | Sales Forecast Accuracy | High | Stable | Repetitive | Low | Very High
SDV10118 | Market Penetration Rate | Critical | Stable | Irregular | Low | Very High
SDV10119 | Revenue Growth from New Products | High | Decreasing | Repetitive | Low | Very High
MSR10101 | Sales Readiness Index | Critical | Stable | Irregular | Low | Very High
MSR10102 | Training Completion Rate | High | Stable | Repetitive | Low | Very High
MSR10103 | Time to Competency | Critical | Increasing | Irregular | Low | Very High
MSR10104 | Sales Enablement Content Utilization | High | Stable | Repetitive | Low | Very High

MSR10105 | Sales Tool Adoption Rate | Critical | Stable | Irregular | Low | Very High
MSR10106 | Lead Response Time | High | Increasing | Repetitive | Low | Very High
MSR10107 | Customer Support Response Time | High | Increasing | Repetitive | Low | Very High
MSR10108 | Sales Support Ticket Resolution Time | High | Increasing | Repetitive | Low | Very High
MSR10109 | Sales Support Satisfaction | High | Decreasing | Repetitive | Low | Very High
MSR10110 | Sales Support Ticket Volume | Critical | Increasing | Irregular | Low | Very High
MSR10111 | Market Readiness Score | Critical | Increasing | Irregular | Low | Very High
MSR10112 | Content Effectiveness | High | Decreasing | Repetitive | Low | Very High
MSR10113 | Onboarding Time for Sales Reps | High | Decreasing | Repetitive | Low | Very High
MSR10114 | Sales Certification Rate | High | Stable | Repetitive | Low | Very High
MSR10115 | Sales Training Effectiveness | High | Stable | Repetitive | Low | Very High
MSR10116 | Knowledge Base Usage | High | Stable | Repetitive | Low | Very High
MSR10117 | Sales Collateral Production Time | High | Decreasing | Repetitive | Low | Very High
MSR10118 | Sales Support Team Productivity | High | Stable | Repetitive | Low | Very High
MSR10119 | Market Feedback Loop Efficiency | High | Stable | Repetitive | Low | Very High
LYP10101 | Customer Retention Rate | High | Stable | Repetitive | Low | Very High
LYP10102 | Loyalty Program Enrollment Rate | High | Decreasing | Repetitive | Low | Very High
LYP10103 | Reward Redemption Rate | High | Increasing | Repetitive | Low | Very High
LYP10104 | Customer Lifetime Value (CLV) | High | Increasing | Repetitive | Low | Very High
LYP10105 | Repeat Purchase Rate | Critical | Increasing | Irregular | Low | Very High
LYP10106 | Net Promoter Score (NPS) | High | Stable | Repetitive | Low | Very High
LYP10107 | Customer Satisfaction (CSAT) | High | Decreasing | Repetitive | Low | Very High
LYP10108 | Loyalty Program Activity Rate | High | Increasing | Repetitive | Low | Very High
LYP10109 | Average Transaction Value | Critical | Stable | Irregular | Low | Very High
LYP10110 | Churn Rate | Critical | Decreasing | Irregular | Low | Very High
LYP10111 | Engagement Rate | Critical | Decreasing | Irregular | Low | Very High
LYP10112 | Referral Rate | High | Stable | Repetitive | Low | Very High
LYP10113 | Customer Feedback Score | High | Stable | Repetitive | Low | Very High
LYP10114 | Cost per Acquisition (CPA) | High | Decreasing | Repetitive | Low | Very High
LYP10115 | ROI of Loyalty Program | High | Stable | Repetitive | Low | Very High
LYP10116 | Loyalty Program Fraud Rate | Critical | Decreasing | Irregular | Low | Very High
LYP10117 | Loyalty Program Tier Upgrades | Critical | Stable | Irregular | Low | Very High
LYP10118 | Inactive Member Rate | Critical | Increasing | Irregular | Low | Very High
LYP10119 | Loyalty Points Expiration Rate | Critical | Stable | Irregular | Low | Very High
CPG10101 | Campaign Reach | High | Increasing | Repetitive | Low | Very High
CPG10102 | Impressions | Critical | Decreasing | Irregular | Low | Very High
CPG10103 | Click-Through Rate (CTR) | High | Decreasing | Repetitive | Low | Very High
CPG10104 | Conversion Rate | Critical | Decreasing | Irregular | Low | Very High
CPG10105 | Lead Generation Rate | Critical | Decreasing | Irregular | Low | Very High
CPG10106 | Cost Per Lead (CPL) | Critical | Decreasing | Irregular | Low | Very High

CPG10107 | Return on Marketing Investment (ROMI) | High | Decreasing | Repetitive | Low | Very High
CPG10108 | Sales Revenue | High | Decreasing | Repetitive | Low | Very High
CPG10109 | Customer Acquisition Cost (CAC) | High | Increasing | Repetitive | Low | Very High
CPG10110 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
CPG10111 | Engagement Rate | High | Decreasing | Repetitive | Low | Very High
CPG10112 | Bounce Rate | High | Decreasing | Repetitive | Low | Very High
CPG10113 | Campaign ROI | Critical | Increasing | Irregular | Low | Very High
CPG10114 | Customer Satisfaction (CSAT) | High | Decreasing | Repetitive | Low | Very High
CPG10115 | Net Promoter Score (NPS) | Critical | Decreasing | Irregular | Low | Very High
CPG10116 | Brand Awareness Increase | High | Stable | Repetitive | Low | Very High
CPG10117 | Social Media Mentions | High | Stable | Repetitive | Low | Very High
CPG10118 | Email Open Rate | Critical | Stable | Irregular | Low | Very High
CPG10119 | Email Click-Through Rate | High | Decreasing | Repetitive | Low | Very High
MKC10101 | Message Reach | Critical | Increasing | Irregular | Low | Very High
MKC10102 | Engagement Rate | Critical | Decreasing | Irregular | Low | Very High
MKC10103 | Click-Through Rate (CTR) | High | Stable | Repetitive | Low | Very High
MKC10104 | Conversion Rate | Critical | Decreasing | Irregular | Low | Very High
MKC10105 | Impressions | Critical | Stable | Irregular | Low | Very High
MKC10106 | Open Rate | High | Increasing | Repetitive | Low | Very High
MKC10107 | Bounce Rate | High | Increasing | Repetitive | Low | Very High
MKC10108 | Social Media Mentions | Critical | Decreasing | Irregular | Low | Very High
MKC10109 | Share of Voice | Critical | Decreasing | Irregular | Low | Very High
MKC10110 | Sentiment Analysis | Critical | Stable | Irregular | Low | Very High
MKC10111 | Brand Awareness Increase | High | Increasing | Repetitive | Low | Very High
MKC10112 | Lead Generation Rate | Critical | Increasing | Irregular | Low | Very High
MKC10113 | Cost Per Click (CPC) | Critical | Stable | Irregular | Low | Very High
MKC10114 | Customer Feedback Score | Critical | Decreasing | Irregular | Low | Very High
MKC10115 | Net Promoter Score (NPS) | High | Increasing | Repetitive | Low | Very High
MKC10116 | Return on Marketing Investment (ROMI) | High | Decreasing | Repetitive | Low | Very High
MKC10117 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
MKC10118 | Time on Page | High | Stable | Repetitive | Low | Very High
MKC10119 | Content Share Rate | High | Stable | Repetitive | Low | Very High
ADS10101 | Reach | High | Stable | Repetitive | Low | Very High
ADS10102 | Impressions | Critical | Increasing | Irregular | Low | Very High
ADS10103 | Click-Through Rate (CTR) | High | Increasing | Repetitive | Low | Very High
ADS10104 | Conversion Rate | Critical | Decreasing | Irregular | Low | Very High
ADS10105 | Cost Per Click (CPC) | Critical | Decreasing | Irregular | Low | Very High
ADS10106 | Cost Per Thousand Impressions (CPM) | Critical | Increasing | Irregular | Low | Very High
ADS10107 | Return on Advertising Spend (ROAS) | High | Increasing | Repetitive | Low | Very High
ADS10108 | Ad Engagement Rate | High | Stable | Repetitive | Low | Very High

ADS10109 | Ad Frequency | High | Decreasing | Repetitive | Low | Very High
ADS10110 | Bounce Rate | High | Increasing | Repetitive | Low | Very High
ADS10111 | Conversion Value | High | Increasing | Repetitive | Low | Very High
ADS10112 | Ad Recall | High | Increasing | Repetitive | Low | Very High
ADS10113 | Customer Acquisition Cost (CAC) | High | Decreasing | Repetitive | Low | Very High
ADS10114 | Lead Generation Rate | Critical | Decreasing | Irregular | Low | Very High
ADS10115 | Ad Sentiment Analysis | Critical | Stable | Irregular | Low | Very High
ADS10116 | Ad Share of Voice | High | Stable | Repetitive | Low | Very High
ADS10117 | Social Media Mentions | High | Increasing | Repetitive | Low | Very High
ADS10118 | Video View Rate | High | Increasing | Repetitive | Low | Very High
ADS10119 | Ad Viewability Rate | High | Increasing | Repetitive | Low | Very High

=== inconsistencies_report_mkts.txt ===

Inconsistencies Analysis Report
Generated on: 2026-09-20 15:00:20

Metrics Dependent Inconsistencies:
None found.

Metrics Independent Inconsistencies:
18336 pair(s) found; showing top 30 by percentage:

Metric 1 ID	Metric 1 Name	Metric 2 ID	Metric 2 Name	Correlation	Count	Percentage	Over Points
LYP10108	Loyalty Program Activity Rate	CPG10106	Cost Per Lead (CPL)	0.04	906	30.20%	3000
SSP10110	Sales Forecast Accuracy	ADS10115	Ad Sentiment Analysis	0.02	904	30.13%	3000
MKR10111	Consumer Awareness and Perception	CPG10104	Conversion Rate	0.03	900	30.00%	3000
MSR10108	Sales Support Ticket Resolution Time	CPG10112	Bounce Rate	0.03	900	30.00%	3000
SSP10108	Sales Growth	SFO10101	Forecast Accuracy	0.02	898	29.93%	3000
BRM10110	Brand Sentiment	ADS10109	Ad Frequency	0.03	898	29.93%	3000
BRM10119	Brand Extension Success	MKR10118	Customer Lifetime Value (CLV)	0.02	896	29.87%	3000
SFO10103	Sales Forecast Variance	LYP10113	Customer Feedback Score	0.03	895	29.83%	3000
SFO10110	Sales Forecasting Methodology	ADS10117	Social Media Mentions	0.03	895	29.83%	3000
MSP10104	Customer Lifetime Value (CLV)	MKC10112	Lead Generation Rate	0.05	894	29.80%	3000
MKR10102	Market Share by Segment	MKR10118	Customer Lifetime Value (CLV)	0.02	894	29.80%	3000
CPG10103	Click-Through Rate (CTR)	MKC10102	Engagement Rate	0.04	894	29.80%	3000
CPG10115	Net Promoter Score (NPS)	ADS10111	Conversion Value	0.02	892	29.73%	3000
MKR10109	Product Performance	CPG10112	Bounce Rate	0.02	891	29.70%	3000
MSP10110	Customer Satisfaction	CHM10106	Customer Acquisition by Channel	0.05	890	29.67%	3000
SFO10106	Conversion Rate by Stage	ADS10104	Conversion Rate	0.03	890	29.67%	3000
MKR10109	Product Performance	LYP10119	Loyalty Points Expiration Rate	0.04	890	29.67%	3000
SDV10102	Lead Generation Rate	CPG10103	Click-Through Rate (CTR)	0.02	890	29.67%	3000

LYP10103 | Reward Redemption Rate | CPG10116 | Brand Awareness Increase | 0.05 | 890 | 29.67% | 3000
LYP10117 | Loyalty Program Tier Upgrades | ADS10110 | Bounce Rate | 0.02 | 890 | 29.67% | 3000
MKR10101 | Market Size and Growth | MSR10115 | Sales Training Effectiveness | 0.04 | 889 | 29.63% | 3000
LYP10112 | Referral Rate | MKC10115 | Net Promoter Score (NPS) | 0.01 | 889 | 29.63% | 3000
MKR10115 | Regulatory Impact | CPG10117 | Social Media Mentions | 0.01 | 888 | 29.60% | 3000
MKR10118 | Customer Lifetime Value (CLV) | ADS10107 | Return on Advertising Spend (ROAS) | 0.05 | 888 | 29.60% | 3000
MKC10110 | Sentiment Analysis | MKC10117 | Customer Retention Rate | 0.01 | 888 | 29.60% | 3000
SDV10101 | Sales Growth Rate | SDV10107 | Sales Team Productivity | 0.03 | 886 | 29.53% | 3000
MSP10103 | Customer Retention Rate | MKR10105 | Competitive Analysis | 0.04 | 885 | 29.50% | 3000
MSR10108 | Sales Support Ticket Resolution Time | MKC10119 | Content Share Rate | 0.02 | 885 | 29.50% | 3000
SFO10106 | Conversion Rate by Stage | ADS10110 | Bounce Rate | -0.00 | 884 | 29.47% | 3000
SDV10109 | Average Revenue per User (ARPU) | SDV10111 | Sales Opportunity Volume | 0.03 | 884 | 29.47% | 3000

Groups Dependent Inconsistencies:

None found.

Groups Independent Inconsistencies:

66 pair(s) found; showing top 30 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

CHM10001 | CHM10001 | MKC10001 | MKC10001 | 0.01 | 876 | 29.20% | 3000
MSP10001 | MSP10001 | SSP10001 | SSP10001 | 0.02 | 872 | 29.07% | 3000
CPG10001 | CPG10001 | ADS10001 | ADS10001 | 0.00 | 867 | 28.90% | 3000
LYP10001 | LYP10001 | MKC10001 | MKC10001 | 0.01 | 866 | 28.87% | 3000
SDV10001 | SDV10001 | ADS10001 | ADS10001 | 0.04 | 865 | 28.83% | 3000
SSP10001 | SSP10001 | ADS10001 | ADS10001 | 0.03 | 859 | 28.63% | 3000
SFO10001 | SFO10001 | MSR10001 | MSR10001 | -0.02 | 858 | 28.60% | 3000
SFO10001 | SFO10001 | CPG10001 | CPG10001 | 0.02 | 854 | 28.47% | 3000
SFO10001 | SFO10001 | SDV10001 | SDV10001 | 0.03 | 852 | 28.40% | 3000
CHM10001 | CHM10001 | SFO10001 | SFO10001 | 0.01 | 848 | 28.27% | 3000
BRM10001 | BRM10001 | LYP10001 | LYP10001 | -0.01 | 844 | 28.13% | 3000
MSR10001 | MSR10001 | LYP10001 | LYP10001 | -0.00 | 842 | 28.07% | 3000
BRM10001 | BRM10001 | SDV10001 | SDV10001 | -0.02 | 841 | 28.03% | 3000
MSP10001 | MSP10001 | ADS10001 | ADS10001 | -0.00 | 840 | 28.00% | 3000
SSP10001 | SSP10001 | SFO10001 | SFO10001 | 0.00 | 838 | 27.93% | 3000
BRM10001 | BRM10001 | MKR10001 | MKR10001 | 0.02 | 838 | 27.93% | 3000
CHM10001 | CHM10001 | SDV10001 | SDV10001 | 0.00 | 836 | 27.87% | 3000
SFO10001 | SFO10001 | ADS10001 | ADS10001 | -0.00 | 836 | 27.87% | 3000
SSP10001 | SSP10001 | BRM10001 | BRM10001 | -0.02 | 833 | 27.77% | 3000
SFO10001 | SFO10001 | MKC10001 | MKC10001 | 0.01 | 833 | 27.77% | 3000

CPG10001 | CPG10001 | MKC10001 | MKC10001 | 0.03 | 833 | 27.77% | 3000
MSP10001 | MSP10001 | SFO10001 | SFO10001 | -0.00 | 830 | 27.67% | 3000
MSR10001 | MSR10001 | ADS10001 | ADS10001 | 0.00 | 829 | 27.63% | 3000
MSP10001 | MSP10001 | LYP10001 | LYP10001 | 0.00 | 828 | 27.60% | 3000
SDV10001 | SDV10001 | MSR10001 | MSR10001 | -0.01 | 828 | 27.60% | 3000
MSR10001 | MSR10001 | CPG10001 | CPG10001 | -0.03 | 828 | 27.60% | 3000
SFO10001 | SFO10001 | LYP10001 | LYP10001 | 0.03 | 826 | 27.53% | 3000
CHM10001 | CHM10001 | MSR10001 | MSR10001 | -0.03 | 825 | 27.50% | 3000
MKR10001 | MKR10001 | MSR10001 | MSR10001 | -0.00 | 825 | 27.50% | 3000
MSP10001 | MSP10001 | MSR10001 | MSR10001 | -0.04 | 822 | 27.40% | 3000

Sub_Modules Dependent Inconsistencies:

None found.

Sub_Modules Independent Inconsistencies:

66 pair(s) found; showing top 30 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

CHM10000 | CHM10000 | MKC10000 | MKC10000 | 0.01 | 876 | 29.20% | 3000
MSP10000 | MSP10000 | SSP10000 | SSP10000 | 0.02 | 872 | 29.07% | 3000
CPG10000 | CPG10000 | ADS10000 | ADS10000 | 0.00 | 867 | 28.90% | 3000
LYP10000 | LYP10000 | MKC10000 | MKC10000 | 0.01 | 866 | 28.87% | 3000
SDV10000 | SDV10000 | ADS10000 | ADS10000 | 0.04 | 865 | 28.83% | 3000
SSP10000 | SSP10000 | ADS10000 | ADS10000 | 0.03 | 859 | 28.63% | 3000
SFO10000 | SFO10000 | MSR10000 | MSR10000 | -0.02 | 858 | 28.60% | 3000
SFO10000 | SFO10000 | CPG10000 | CPG10000 | 0.02 | 854 | 28.47% | 3000
SFO10000 | SFO10000 | SDV10000 | SDV10000 | 0.03 | 852 | 28.40% | 3000
CHM10000 | CHM10000 | SFO10000 | SFO10000 | 0.01 | 848 | 28.27% | 3000
BRM10000 | BRM10000 | LYP10000 | LYP10000 | -0.01 | 844 | 28.13% | 3000
MSR10000 | MSR10000 | LYP10000 | LYP10000 | -0.00 | 842 | 28.07% | 3000
BRM10000 | BRM10000 | SDV10000 | SDV10000 | -0.02 | 841 | 28.03% | 3000
MSP10000 | MSP10000 | ADS10000 | ADS10000 | -0.00 | 840 | 28.00% | 3000
SSP10000 | SSP10000 | SFO10000 | SFO10000 | 0.00 | 838 | 27.93% | 3000
BRM10000 | BRM10000 | MKR10000 | MKR10000 | 0.02 | 838 | 27.93% | 3000
CHM10000 | CHM10000 | SDV10000 | SDV10000 | 0.00 | 836 | 27.87% | 3000
SFO10000 | SFO10000 | ADS10000 | ADS10000 | -0.00 | 836 | 27.87% | 3000
SSP10000 | SSP10000 | BRM10000 | BRM10000 | -0.02 | 833 | 27.77% | 3000
SFO10000 | SFO10000 | MKC10000 | MKC10000 | 0.01 | 833 | 27.77% | 3000
CPG10000 | CPG10000 | MKC10000 | MKC10000 | 0.03 | 833 | 27.77% | 3000
MSP10000 | MSP10000 | SFO10000 | SFO10000 | -0.00 | 830 | 27.67% | 3000
MSR10000 | MSR10000 | ADS10000 | ADS10000 | 0.00 | 829 | 27.63% | 3000
MSP10000 | MSP10000 | LYP10000 | LYP10000 | 0.00 | 828 | 27.60% | 3000

SDV10000 | SDV10000 | MSR10000 | MSR10000 | -0.01 | 828 | 27.60% | 3000
MSR10000 | MSR10000 | CPG10000 | CPG10000 | -0.03 | 828 | 27.60% | 3000
SFO10000 | SFO10000 | LYP10000 | LYP10000 | 0.03 | 826 | 27.53% | 3000
CHM10000 | CHM10000 | MSR10000 | MSR10000 | -0.03 | 825 | 27.50% | 3000
MKR10000 | MKR10000 | MSR10000 | MSR10000 | -0.00 | 825 | 27.50% | 3000
MSP10000 | MSP10000 | MSR10000 | MSR10000 | -0.04 | 822 | 27.40% | 3000

=== trends_and_repetitions_report_mkts.txt ===

Metrics Trends:

MSP10101: Decreasing

MSP10102: Decreasing

MSP10103: Stable

MSP10104: Increasing

MSP10105: Decreasing

MSP10106: Decreasing

MSP10107: Decreasing

MSP10108: Decreasing

MSP10109: Decreasing

MSP10110: Decreasing

SSP10101: Stable

SSP10102: Stable

SSP10103: Stable

SSP10104: Stable

SSP10105: Stable

SSP10106: Increasing

SSP10107: Decreasing

SSP10108: Increasing

SSP10109: Increasing

SSP10110: Decreasing

CHM10101: Stable

CHM10102: Decreasing

CHM10103: Stable

CHM10104: Decreasing

CHM10105: Decreasing

CHM10106: Decreasing

CHM10107: Decreasing

CHM10108: Decreasing

CHM10109: Stable

CHM10110: Stable

SFO10101: Stable
SFO10102: Increasing
SFO10103: Decreasing
SFO10104: Decreasing
SFO10105: Decreasing
SFO10106: Stable
SFO10107: Stable
SFO10108: Decreasing
SFO10109: Stable
SFO10110: Increasing
BRM10101: Stable
BRM10102: Decreasing
BRM10103: Increasing
BRM10104: Increasing
BRM10105: Stable
BRM10106: Decreasing
BRM10107: Increasing
BRM10108: Increasing
BRM10109: Stable
BRM10110: Decreasing
BRM10111: Increasing
BRM10112: Stable
BRM10113: Increasing
BRM10114: Stable
BRM10115: Decreasing
BRM10116: Decreasing
BRM10117: Stable
BRM10118: Increasing
BRM10119: Increasing
MKR10101: Increasing
MKR10102: Stable
MKR10103: Decreasing
MKR10104: Stable
MKR10105: Decreasing
MKR10106: Stable
MKR10107: Stable
MKR10108: Increasing
MKR10109: Increasing
MKR10110: Stable
MKR10111: Decreasing

MKR10112: Stable
MKR10113: Stable
MKR10114: Increasing
MKR10115: Decreasing
MKR10116: Increasing
MKR10117: Stable
MKR10118: Stable
MKR10119: Decreasing
SDV10101: Stable
SDV10102: Decreasing
SDV10103: Stable
SDV10104: Decreasing
SDV10105: Increasing
SDV10106: Stable
SDV10107: Decreasing
SDV10108: Increasing
SDV10109: Stable
SDV10110: Stable
SDV10111: Decreasing
SDV10112: Decreasing
SDV10113: Stable
SDV10114: Decreasing
SDV10115: Decreasing
SDV10116: Decreasing
SDV10117: Stable
SDV10118: Stable
SDV10119: Decreasing
MSR10101: Stable
MSR10102: Stable
MSR10103: Increasing
MSR10104: Stable
MSR10105: Stable
MSR10106: Increasing
MSR10107: Increasing
MSR10108: Increasing
MSR10109: Decreasing
MSR10110: Increasing
MSR10111: Increasing
MSR10112: Decreasing
MSR10113: Decreasing

MSR10114: Stable
MSR10115: Stable
MSR10116: Stable
MSR10117: Decreasing
MSR10118: Stable
MSR10119: Stable
LYP10101: Stable
LYP10102: Decreasing
LYP10103: Increasing
LYP10104: Increasing
LYP10105: Increasing
LYP10106: Stable
LYP10107: Decreasing
LYP10108: Increasing
LYP10109: Stable
LYP10110: Decreasing
LYP10111: Decreasing
LYP10112: Stable
LYP10113: Stable
LYP10114: Decreasing
LYP10115: Stable
LYP10116: Decreasing
LYP10117: Stable
LYP10118: Increasing
LYP10119: Stable
CPG10101: Increasing
CPG10102: Decreasing
CPG10103: Decreasing
CPG10104: Decreasing
CPG10105: Decreasing
CPG10106: Decreasing
CPG10107: Decreasing
CPG10108: Decreasing
CPG10109: Increasing
CPG10110: Stable
CPG10111: Decreasing
CPG10112: Decreasing
CPG10113: Increasing
CPG10114: Decreasing
CPG10115: Decreasing

CPG10116: Stable
CPG10117: Stable
CPG10118: Stable
CPG10119: Decreasing
MKC10101: Increasing
MKC10102: Decreasing
MKC10103: Stable
MKC10104: Decreasing
MKC10105: Stable
MKC10106: Increasing
MKC10107: Increasing
MKC10108: Decreasing
MKC10109: Decreasing
MKC10110: Stable
MKC10111: Increasing
MKC10112: Increasing
MKC10113: Stable
MKC10114: Decreasing
MKC10115: Increasing
MKC10116: Decreasing
MKC10117: Stable
MKC10118: Stable
MKC10119: Stable
ADS10101: Stable
ADS10102: Increasing
ADS10103: Increasing
ADS10104: Decreasing
ADS10105: Decreasing
ADS10106: Increasing
ADS10107: Increasing
ADS10108: Stable
ADS10109: Decreasing
ADS10110: Increasing
ADS10111: Increasing
ADS10112: Increasing
ADS10113: Decreasing
ADS10114: Decreasing
ADS10115: Stable
ADS10116: Stable
ADS10117: Increasing

ADS10118: Increasing

ADS10119: Increasing

Metrics Repetitions (Fourier Analysis):

MSP10101: No clear repetition (potential instability)

MSP10102: Dominant period: 3.43 units (repetitive)

MSP10103: Dominant period: 2.06 units (repetitive)

MSP10104: Dominant period: 13.22 units (repetitive)

MSP10105: Dominant period: 2.36 units (repetitive)

MSP10106: Dominant period: 49.18 units (repetitive)

MSP10107: Dominant period: 11.11 units (repetitive)

MSP10108: No clear repetition (potential instability)

MSP10109: Dominant period: 2.01 units (repetitive)

MSP10110: Dominant period: 2.25 units (repetitive)

SSP10101: Dominant period: 3.34 units (repetitive)

SSP10102: No clear repetition (potential instability)

SSP10103: No clear repetition (potential instability)

SSP10104: Dominant period: 6.70 units (repetitive)

SSP10105: No clear repetition (potential instability)

SSP10106: Dominant period: 8.26 units (repetitive)

SSP10107: No clear repetition (potential instability)

SSP10108: Dominant period: 4.46 units (repetitive)

SSP10109: Dominant period: 9.74 units (repetitive)

SSP10110: Dominant period: 3.90 units (repetitive)

CHM10101: Dominant period: 7.98 units (repetitive)

CHM10102: Dominant period: 5.24 units (repetitive)

CHM10103: Dominant period: 5.19 units (repetitive)

CHM10104: Dominant period: 5.20 units (repetitive)

CHM10105: Dominant period: 3.08 units (repetitive)

CHM10106: Dominant period: 4.22 units (repetitive)

CHM10107: No clear repetition (potential instability)

CHM10108: No clear repetition (potential instability)

CHM10109: Dominant period: 3.42 units (repetitive)

CHM10110: No clear repetition (potential instability)

SFO10101: Dominant period: 2.66 units (repetitive)

SFO10102: No clear repetition (potential instability)

SFO10103: Dominant period: 2.61 units (repetitive)

SFO10104: No clear repetition (potential instability)

SFO10105: No clear repetition (potential instability)

SFO10106: No clear repetition (potential instability)

SFO10107: Dominant period: 3.74 units (repetitive)
SFO10108: Dominant period: 20.98 units (repetitive)
SFO10109: Dominant period: 2.20 units (repetitive)
SFO10110: Dominant period: 4.42 units (repetitive)
BRM10101: No clear repetition (potential instability)
BRM10102: No clear repetition (potential instability)
BRM10103: No clear repetition (potential instability)
BRM10104: Dominant period: 103.45 units (repetitive)
BRM10105: Dominant period: 5.96 units (repetitive)
BRM10106: Dominant period: 19.87 units (repetitive)
BRM10107: Dominant period: 9.90 units (repetitive)
BRM10108: Dominant period: 3.12 units (repetitive)
BRM10109: Dominant period: 4.66 units (repetitive)
BRM10110: Dominant period: 17.34 units (repetitive)
BRM10111: Dominant period: 15.23 units (repetitive)
BRM10112: Dominant period: 15.71 units (repetitive)
BRM10113: Dominant period: 8.11 units (repetitive)
BRM10114: Dominant period: 6.98 units (repetitive)
BRM10115: Dominant period: 2.06 units (repetitive)
BRM10116: No clear repetition (potential instability)
BRM10117: No clear repetition (potential instability)
BRM10118: Dominant period: 6.06 units (repetitive)
BRM10119: No clear repetition (potential instability)
MKR10101: Dominant period: 10.31 units (repetitive)
MKR10102: No clear repetition (potential instability)
MKR10103: Dominant period: 3.12 units (repetitive)
MKR10104: Dominant period: 11.54 units (repetitive)
MKR10105: Dominant period: 6.68 units (repetitive)
MKR10106: No clear repetition (potential instability)
MKR10107: No clear repetition (potential instability)
MKR10108: No clear repetition (potential instability)
MKR10109: Dominant period: 3.01 units (repetitive)
MKR10110: Dominant period: 2.25 units (repetitive)
MKR10111: Dominant period: 2.97 units (repetitive)
MKR10112: Dominant period: 4.14 units (repetitive)
MKR10113: No clear repetition (potential instability)
MKR10114: No clear repetition (potential instability)
MKR10115: Dominant period: 2.91 units (repetitive)
MKR10116: No clear repetition (potential instability)
MKR10117: No clear repetition (potential instability)

MKR10118: No clear repetition (potential instability)
MKR10119: No clear repetition (potential instability)
SDV10101: No clear repetition (potential instability)
SDV10102: No clear repetition (potential instability)
SDV10103: Dominant period: 20.69 units (repetitive)
SDV10104: No clear repetition (potential instability)
SDV10105: No clear repetition (potential instability)
SDV10106: No clear repetition (potential instability)
SDV10107: Dominant period: 11.28 units (repetitive)
SDV10108: No clear repetition (potential instability)
SDV10109: No clear repetition (potential instability)
SDV10110: Dominant period: 8.29 units (repetitive)
SDV10111: No clear repetition (potential instability)
SDV10112: Dominant period: 65.22 units (repetitive)
SDV10113: No clear repetition (potential instability)
SDV10114: Dominant period: 10.87 units (repetitive)
SDV10115: No clear repetition (potential instability)
SDV10116: No clear repetition (potential instability)
SDV10117: Dominant period: 9.15 units (repetitive)
SDV10118: No clear repetition (potential instability)
SDV10119: Dominant period: 10.00 units (repetitive)
MSR10101: No clear repetition (potential instability)
MSR10102: Dominant period: 9.09 units (repetitive)
MSR10103: No clear repetition (potential instability)
MSR10104: Dominant period: 2.71 units (repetitive)
MSR10105: No clear repetition (potential instability)
MSR10106: Dominant period: 2.03 units (repetitive)
MSR10107: Dominant period: 3.29 units (repetitive)
MSR10108: Dominant period: 2.38 units (repetitive)
MSR10109: Dominant period: 12.61 units (repetitive)
MSR10110: No clear repetition (potential instability)
MSR10111: No clear repetition (potential instability)
MSR10112: Dominant period: 8.52 units (repetitive)
MSR10113: Dominant period: 12.45 units (repetitive)
MSR10114: Dominant period: 4.03 units (repetitive)
MSR10115: Dominant period: 17.05 units (repetitive)
MSR10116: Dominant period: 2.84 units (repetitive)
MSR10117: Dominant period: 2.38 units (repetitive)
MSR10118: Dominant period: 5.47 units (repetitive)
MSR10119: Dominant period: 2.08 units (repetitive)

LYP10101: Dominant period: 4.34 units (repetitive)
LYP10102: Dominant period: 4.78 units (repetitive)
LYP10103: Dominant period: 7.89 units (repetitive)
LYP10104: Dominant period: 2.58 units (repetitive)
LYP10105: No clear repetition (potential instability)
LYP10106: Dominant period: 4.70 units (repetitive)
LYP10107: Dominant period: 6.62 units (repetitive)
LYP10108: Dominant period: 2.01 units (repetitive)
LYP10109: No clear repetition (potential instability)
LYP10110: No clear repetition (potential instability)
LYP10111: No clear repetition (potential instability)
LYP10112: Dominant period: 2.60 units (repetitive)
LYP10113: Dominant period: 7.50 units (repetitive)
LYP10114: Dominant period: 2.67 units (repetitive)
LYP10115: Dominant period: 28.57 units (repetitive)
LYP10116: No clear repetition (potential instability)
LYP10117: No clear repetition (potential instability)
LYP10118: No clear repetition (potential instability)
LYP10119: No clear repetition (potential instability)
CPG10101: Dominant period: 10.45 units (repetitive)
CPG10102: No clear repetition (potential instability)
CPG10103: Dominant period: 12.10 units (repetitive)
CPG10104: No clear repetition (potential instability)
CPG10105: No clear repetition (potential instability)
CPG10106: No clear repetition (potential instability)
CPG10107: Dominant period: 62.50 units (repetitive)
CPG10108: Dominant period: 4.10 units (repetitive)
CPG10109: Dominant period: 4.96 units (repetitive)
CPG10110: No clear repetition (potential instability)
CPG10111: Dominant period: 3.45 units (repetitive)
CPG10112: Dominant period: 8.88 units (repetitive)
CPG10113: No clear repetition (potential instability)
CPG10114: Dominant period: 4.04 units (repetitive)
CPG10115: No clear repetition (potential instability)
CPG10116: Dominant period: 2.31 units (repetitive)
CPG10117: Dominant period: 2.09 units (repetitive)
CPG10118: No clear repetition (potential instability)
CPG10119: Dominant period: 2.42 units (repetitive)
MKC10101: No clear repetition (potential instability)
MKC10102: No clear repetition (potential instability)

MKC10103: Dominant period: 17.14 units (repetitive)
MKC10104: No clear repetition (potential instability)
MKC10105: No clear repetition (potential instability)
MKC10106: Dominant period: 15.54 units (repetitive)
MKC10107: Dominant period: 4.27 units (repetitive)
MKC10108: No clear repetition (potential instability)
MKC10109: No clear repetition (potential instability)
MKC10110: No clear repetition (potential instability)
MKC10111: Dominant period: 166.67 units (repetitive)
MKC10112: No clear repetition (potential instability)
MKC10113: No clear repetition (potential instability)
MKC10114: No clear repetition (potential instability)
MKC10115: Dominant period: 750.00 units (repetitive)
MKC10116: Dominant period: 19.48 units (repetitive)
MKC10117: No clear repetition (potential instability)
MKC10118: Dominant period: 14.42 units (repetitive)
MKC10119: Dominant period: 250.00 units (repetitive)
ADS10101: Dominant period: 10.45 units (repetitive)
ADS10102: No clear repetition (potential instability)
ADS10103: Dominant period: 8.36 units (repetitive)
ADS10104: No clear repetition (potential instability)
ADS10105: No clear repetition (potential instability)
ADS10106: No clear repetition (potential instability)
ADS10107: Dominant period: 6.93 units (repetitive)
ADS10108: Dominant period: 8.55 units (repetitive)
ADS10109: Dominant period: 6.30 units (repetitive)
ADS10110: Dominant period: 2.40 units (repetitive)
ADS10111: Dominant period: 12.00 units (repetitive)
ADS10112: Dominant period: 13.82 units (repetitive)
ADS10113: Dominant period: 6.28 units (repetitive)
ADS10114: No clear repetition (potential instability)
ADS10115: No clear repetition (potential instability)
ADS10116: Dominant period: 8.88 units (repetitive)
ADS10117: Dominant period: 49.18 units (repetitive)
ADS10118: Dominant period: 2.41 units (repetitive)
ADS10119: Dominant period: 2.50 units (repetitive)

Groups Trends:

MSP10001: Decreasing

SSP10001: Stable

CHM10001: Decreasing
SFO10001: Decreasing
BRM10001: Stable
MKR10001: Stable
SDV10001: Decreasing
MSR10001: Stable
LYP10001: Decreasing
CPG10001: Decreasing
MKC10001: Decreasing
ADS10001: Increasing

Groups Repetitions (Fourier Analysis):

MSP10001: Dominant period: 750.00 units (repetitive)
SSP10001: Dominant period: 6.10 units (repetitive)
CHM10001: Dominant period: 2.35 units (repetitive)
SFO10001: Dominant period: 2.30 units (repetitive)
BRM10001: Dominant period: 6.45 units (repetitive)
MKR10001: No clear repetition (potential instability)
SDV10001: Dominant period: 3.01 units (repetitive)
MSR10001: Dominant period: 26.32 units (repetitive)
LYP10001: Dominant period: 16.95 units (repetitive)
CPG10001: No clear repetition (potential instability)
MKC10001: Dominant period: 3.67 units (repetitive)
ADS10001: Dominant period: 3.18 units (repetitive)

Sub_Modules Trends:

MSP10000: Decreasing
SSP10000: Stable
CHM10000: Decreasing
SFO10000: Decreasing
BRM10000: Stable
MKR10000: Stable
SDV10000: Decreasing
MSR10000: Stable
LYP10000: Decreasing
CPG10000: Decreasing
MKC10000: Decreasing
ADS10000: Increasing

Sub_Modules Repetitions (Fourier Analysis):

MSP10000: Dominant period: 750.00 units (repetitive)
SSP10000: Dominant period: 6.10 units (repetitive)

CHM10000: Dominant period: 2.35 units (repetitive)
 SFO10000: Dominant period: 2.30 units (repetitive)
 BRM10000: Dominant period: 6.45 units (repetitive)
 MKR10000: No clear repetition (potential instability)
 SDV10000: Dominant period: 3.01 units (repetitive)
 MSR10000: Dominant period: 26.32 units (repetitive)
 LYP10000: Dominant period: 16.95 units (repetitive)
 CPG10000: No clear repetition (potential instability)
 MKC10000: Dominant period: 3.67 units (repetitive)
 ADS10000: Dominant period: 3.18 units (repetitive)

=== metrics_summary_mkts.txt ===

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
ADS10101	45.12	14.85	89.68	0.57	0	35	2026-01-07
ADS10102	44.54	14.88	89.18	-0.09	0	35	2026-01-07
ADS10103	44.68	15.19	90.23	-0.88	0	35	2026-01-07
ADS10104	45.03	14.61	88.87	1.19	0	35	2026-01-07
ADS10105	45.22	15.08	90.44	-0.01	0	35	2026-01-07
ADS10106	45.1	14.99	90.07	0.13	0	35	2026-01-07
ADS10107	44.81	15.29	90.67	-1.04	0	35	2026-01-07
ADS10108	44.6	15.0	89.6	-0.4	0	35	2026-01-07
ADS10109	45.35	14.92	90.09	0.6	0	35	2026-01-07
ADS10110	44.99	15.01	90.01	-0.03	0	35	2026-01-07
ADS10111	45.28	15.0	90.29	0.27	0	35	2026-01-07
ADS10112	45.12	14.79	89.5	0.74	0	35	2026-01-07
ADS10113	45.01	15.36	91.08	-1.06	0	35	2026-01-07
ADS10114	45.01	15.23	90.7	-0.68	0	35	2026-01-07
ADS10115	45.2	15.13	90.59	-0.19	0	35	2026-01-07
ADS10116	45.02	15.02	90.07	-0.02	0	35	2026-01-07
ADS10117	44.81	15.16	90.28	-0.66	0	35	2026-01-07
ADS10118	44.93	15.2	90.53	-0.68	0	35	2026-01-07
ADS10119	44.95	15.22	90.61	-0.7	0	35	2026-01-07
BRM10101	44.82	15.15	90.29	-0.64	0	35	2026-01-07
BRM10102	44.81	15.0	89.8	-0.19	0	35	2026-01-07
BRM10103	45.47	15.0	90.46	0.47	0	35	2026-01-07
BRM10104	44.87	15.05	90.02	-0.27	0	35	2026-01-07
BRM10105	44.5	14.8	88.91	0.09	0	35	2026-01-07
BRM10106	44.21	14.95	89.06	-0.64	0	35	2026-01-07
BRM10107	45.32	14.95	90.16	0.48	0	35	2026-01-07
BRM10108	44.85	15.35	90.9	-1.19	0	35	2026-01-07
BRM10109	44.54	14.89	89.21	-0.14	0	35	2026-01-07
BRM10110	45.02	14.9	89.71	0.33	0	35	2026-01-07
BRM10111	44.25	15.12	89.59	-1.1	0	35	2026-01-07
BRM10112	45.13	15.0	90.14	0.13	0	35	2026-01-07
BRM10113	45.16	15.26	90.95	-0.62	0	35	2026-01-07
BRM10114	44.78	15.32	90.74	-1.19	0	35	2026-01-07
BRM10115	45.34	15.13	90.74	-0.05	0	35	2026-01-07
BRM10116	44.18	15.06	89.35	-0.99	0	35	2026-01-07
BRM10117	45.27	14.78	89.6	0.93	0	35	2026-01-07
BRM10118	44.67	15.32	90.63	-1.28	0	35	2026-01-07
BRM10119	45.06	14.79	89.43	0.7	0	35	2026-01-07
CHM10101	44.76	15.3	90.65	-1.13	0	35	2026-01-07
CHM10102	44.92	14.8	89.31	0.52	0	35	2026-01-07
CHM10103	44.25	14.97	89.16	-0.66	0	35	2026-01-07
CHM10104	44.58	15.43	90.88	-1.71	0	35	2026-01-07
CHM10105	45.23	15.45	91.59	-1.13	0	35	2026-01-07
CHM10106	44.87	14.95	89.74	0.01	0	35	2026-01-07
CHM10107	45.08	15.16	90.54	-0.39	0	35	2026-01-07
CHM10108	44.85	15.06	90.03	-0.32	0	35	2026-01-07
CHM10109	45.32	15.01	90.35	0.29	0	35	2026-01-07
CHM10110	44.6	15.23	90.29	-1.09	0	35	2026-01-07
CPG10101	45.3	15.18	90.85	-0.25	0	35	2026-01-07
CPG10102	44.95	14.65	88.9	1.0	0	35	2026-01-07
CPG10103	45.18	14.94	89.99	0.37	0	35	2026-01-07

CPG10104	44.85	15.13	90.24	-0.54	0	35	2026-01-07
CPG10105	45.09	15.14	90.5	-0.33	0	35	2026-01-07
CPG10106	45.1	14.93	89.89	0.31	0	35	2026-01-07
CPG10107	45.05	14.9	89.74	0.35	0	35	2026-01-07
CPG10108	44.83	14.83	89.31	0.34	0	35	2026-01-07
CPG10109	44.84	14.99	89.81	-0.13	0	35	2026-01-07
CPG10110	45.14	15.0	90.15	0.13	0	35	2026-01-07
CPG10111	44.56	14.7	88.66	0.46	0	35	2026-01-07
CPG10112	44.93	15.05	90.07	-0.21	0	35	2026-01-07
CPG10113	44.8	14.95	89.63	-0.04	0	35	2026-01-07
CPG10114	44.92	15.42	91.18	-1.33	0	35	2026-01-07
CPG10115	45.25	14.6	89.06	1.45	0	35	2026-01-07
CPG10116	45.24	15.3	91.15	-0.68	0	35	2026-01-07
CPG10117	45.35	14.72	89.52	1.17	0	35	2026-01-07
CPG10118	45.05	14.86	89.62	0.47	0	35	2026-01-07
CPG10119	44.98	15.0	89.98	-0.01	0	35	2026-01-07
LYP10101	45.06	15.11	90.4	-0.28	0	35	2026-01-07
LYP10102	44.72	14.98	89.66	-0.22	0	35	2026-01-07
LYP10103	45.19	15.27	90.99	-0.61	0	35	2026-01-07
LYP10104	44.99	15.5	91.49	-1.51	0	35	2026-01-07
LYP10105	45.04	15.36	91.12	-1.03	0	35	2026-01-07
LYP10106	45.08	14.92	89.84	0.32	0	35	2026-01-07
LYP10107	44.69	15.34	90.73	-1.34	0	35	2026-01-07
LYP10108	45.14	14.99	90.1	0.18	0	35	2026-01-07
LYP10109	45.22	14.93	90.02	0.42	0	35	2026-01-07
LYP10110	45.18	15.1	90.47	-0.11	0	35	2026-01-07
LYP10111	45.23	15.52	91.79	-1.33	0	35	2026-01-07
LYP10112	45.19	15.0	90.2	0.18	0	35	2026-01-07
LYP10113	44.74	14.98	89.68	-0.19	0	35	2026-01-07
LYP10114	44.31	15.51	90.83	-2.21	0	35	2026-01-07
LYP10115	45.48	14.87	90.1	0.86	0	35	2026-01-07
LYP10116	44.85	14.88	89.5	0.2	0	35	2026-01-07
LYP10117	45.02	14.78	89.37	0.66	0	35	2026-01-07
LYP10118	45.14	15.2	90.74	-0.47	0	35	2026-01-07
LYP10119	44.92	14.89	89.59	0.26	0	35	2026-01-07
MKC10101	45.05	14.89	89.71	0.39	0	35	2026-01-07
MKC10102	45.27	15.25	91.03	-0.48	0	35	2026-01-07
MKC10103	44.89	14.91	89.63	0.15	0	35	2026-01-07
MKC10104	45.08	15.09	90.35	-0.18	0	35	2026-01-07
MKC10105	45.12	15.07	90.32	-0.09	0	35	2026-01-07
MKC10106	45.43	15.13	90.83	0.03	0	35	2026-01-07
MKC10107	45.12	14.98	90.06	0.18	0	35	2026-01-07
MKC10108	44.74	15.12	90.09	-0.61	0	35	2026-01-07
MKC10109	45.59	15.23	91.29	-0.11	0	35	2026-01-07
MKC10110	44.99	14.85	89.55	0.43	0	35	2026-01-07
MKC10111	45.09	15.19	90.67	-0.49	0	35	2026-01-07
MKC10112	45.02	14.8	89.43	0.61	0	35	2026-01-07
MKC10113	45.08	15.44	91.4	-1.24	0	35	2026-01-07
MKC10114	44.99	14.91	89.73	0.25	0	35	2026-01-07
MKC10115	44.95	14.9	89.65	0.25	0	35	2026-01-07
MKC10116	45.19	14.86	89.76	0.62	0	35	2026-01-07
MKC10117	44.89	14.93	89.67	0.11	0	35	2026-01-07
MKC10118	44.38	14.6	88.18	0.57	0	35	2026-01-07
MKC10119	45.04	15.13	90.43	-0.35	0	35	2026-01-07
MKR10101	44.95	14.72	89.11	0.8	0	35	2026-01-07
MKR10102	44.93	15.09	90.2	-0.34	0	35	2026-01-07
MKR10103	45.2	14.83	89.7	0.71	0	35	2026-01-07
MKR10104	45.2	15.01	90.23	0.16	0	35	2026-01-07
MKR10105	44.97	15.24	90.68	-0.74	0	35	2026-01-07
MKR10106	45.41	14.95	90.26	0.55	0	35	2026-01-07
MKR10107	44.73	14.99	89.71	-0.25	0	35	2026-01-07
MKR10108	45.01	15.3	90.93	-0.9	0	35	2026-01-07
MKR10109	44.94	14.78	89.29	0.59	0	35	2026-01-07
MKR10110	45.09	15.23	90.79	-0.61	0	35	2026-01-07
MKR10111	44.98	14.87	89.58	0.38	0	35	2026-01-07
MKR10112	45.1	15.15	90.54	-0.34	0	35	2026-01-07
MKR10113	45.16	15.09	90.43	-0.12	0	35	2026-01-07
MKR10114	44.55	14.82	89.0	0.1	0	35	2026-01-07
MKR10115	45.31	15.13	90.69	-0.07	0	35	2026-01-07
MKR10116	44.56	14.9	89.26	-0.15	0	35	2026-01-07
MKR10117	44.66	15.08	89.9	-0.58	0	35	2026-01-07
MKR10118	44.96	14.89	89.63	0.29	0	35	2026-01-07
MKR10119	44.78	14.93	89.58	-0.02	0	35	2026-01-07
MSP10101	44.89	14.96	89.77	0.01	0	35	2026-01-07
MSP10102	44.66	15.58	91.41	-2.09	0	35	2026-01-07

MSP10103	44.66	15.04	89.77	-0.46	0	35	2026-01-07
MSP10104	45.29	14.82	89.75	0.83	0	35	2026-01-07
MSP10105	44.95	15.36	91.02	-1.11	0	35	2026-01-07
MSP10106	45.24	15.02	90.3	0.18	0	35	2026-01-07
MSP10107	45.35	14.95	90.21	0.49	0	35	2026-01-07
MSP10108	45.0	15.64	91.9	-1.91	0	35	2026-01-07
MSP10109	45.11	15.09	90.38	-0.16	0	35	2026-01-07
MSP10110	45.38	14.6	89.16	1.59	0	35	2026-01-07
MSR10101	45.09	15.34	91.1	-0.92	0	35	2026-01-07
MSR10102	44.58	15.49	91.06	-1.9	0	35	2026-01-07
MSR10103	44.9	15.06	90.09	-0.29	0	35	2026-01-07
MSR10104	44.96	15.02	90.03	-0.11	0	35	2026-01-07
MSR10105	45.18	15.21	90.81	-0.46	0	35	2026-01-07
MSR10106	44.94	15.39	91.12	-1.24	0	35	2026-01-07
MSR10107	44.9	15.09	90.16	-0.37	0	35	2026-01-07
MSR10108	44.86	15.32	90.82	-1.1	0	35	2026-01-07
MSR10109	45.09	14.53	88.67	1.5	0	35	2026-01-07
MSR10110	45.09	15.6	91.89	-1.71	0	35	2026-01-07
MSR10111	45.02	15.04	90.15	-0.1	0	35	2026-01-07
MSR10112	45.16	14.85	89.71	0.61	0	35	2026-01-07
MSR10113	45.21	14.76	89.49	0.93	0	35	2026-01-07
MSR10114	44.72	15.08	89.97	-0.52	0	35	2026-01-07
MSR10115	45.0	14.82	89.47	0.52	0	35	2026-01-07
MSR10116	44.65	15.42	90.92	-1.62	0	35	2026-01-07
MSR10117	44.98	15.26	90.76	-0.8	0	35	2026-01-07
MSR10118	45.21	15.27	91.01	-0.58	0	35	2026-01-07
MSR10119	44.96	15.23	90.66	-0.74	0	35	2026-01-07
SDV10101	44.87	15.17	90.38	-0.63	0	35	2026-01-07
SDV10102	45.04	15.02	90.1	-0.02	0	35	2026-01-07
SDV10103	44.81	15.17	90.31	-0.69	0	35	2026-01-07
SDV10104	45.09	15.32	91.05	-0.86	0	35	2026-01-07
SDV10105	44.86	14.81	89.3	0.41	0	35	2026-01-07
SDV10106	45.51	15.04	90.61	0.4	0	35	2026-01-07
SDV10107	44.63	15.1	89.93	-0.67	0	35	2026-01-07
SDV10108	45.13	15.3	91.02	-0.77	0	35	2026-01-07
SDV10109	45.11	15.2	90.73	-0.5	0	35	2026-01-07
SDV10110	44.81	14.94	89.64	-0.02	0	35	2026-01-07
SDV10111	45.12	15.34	91.15	-0.9	0	35	2026-01-07
SDV10112	45.09	15.11	90.42	-0.24	0	35	2026-01-07
SDV10113	44.78	15.23	90.47	-0.92	0	35	2026-01-07
SDV10114	45.12	15.1	90.43	-0.2	0	35	2026-01-07
SDV10115	45.18	14.85	89.74	0.63	0	35	2026-01-07
SDV10116	44.76	15.21	90.39	-0.87	0	35	2026-01-07
SDV10117	45.21	15.22	90.86	-0.43	0	35	2026-01-07
SDV10118	45.1	14.97	90.01	0.19	0	35	2026-01-07
SDV10119	45.17	15.05	90.31	0.03	0	35	2026-01-07
SF010101	44.8	14.86	89.39	0.22	0	35	2026-01-07
SF010102	44.8	14.99	89.77	-0.17	0	35	2026-01-07
SF010103	44.92	14.9	89.61	0.22	0	35	2026-01-07
SF010104	44.99	14.89	89.66	0.32	0	35	2026-01-07
SF010105	44.79	15.09	90.07	-0.48	0	35	2026-01-07
SF010106	44.82	15.01	89.86	-0.21	0	35	2026-01-07
SF010107	44.44	14.83	88.94	-0.06	0	35	2026-01-07
SF010108	45.12	15.96	92.98	-2.75	0	35	2026-01-07
SF010109	45.22	15.25	90.97	-0.54	0	35	2026-01-07
SF010110	44.89	15.14	90.32	-0.54	0	35	2026-01-07
SSP10101	44.62	15.24	90.35	-1.11	0	35	2026-01-07
SSP10102	45.19	14.93	89.97	0.42	0	35	2026-01-07
SSP10103	44.9	14.97	89.82	-0.03	0	35	2026-01-07
SSP10104	44.81	14.83	89.3	0.31	0	35	2026-01-07
SSP10105	45.08	15.14	90.5	-0.34	0	35	2026-01-07
SSP10106	45.16	14.75	89.4	0.91	0	35	2026-01-07
SSP10107	45.48	15.34	91.5	-0.55	0	35	2026-01-07
SSP10108	44.67	15.0	89.66	-0.33	0	35	2026-01-07
SSP10109	45.82	15.18	91.35	0.29	0	35	2026-01-07
SSP10110	44.96	14.99	89.92	-0.0	0	35	2026-01-07

=== rp_n_summary_mkts.txt ===

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
ADS10101	45.12	14.85	89.68	0.57	0	35	2026-01-07
ADS10102	44.54	14.88	89.18	-0.09	0	35	2026-01-07
ADS10103	44.68	15.19	90.23	-0.88	0	35	2026-01-07

ADS10104	45.03	14.61	88.87	1.19	0	35	2026-01-07
ADS10105	45.22	15.08	90.44	-0.01	0	35	2026-01-07
ADS10106	45.1	14.99	90.07	0.13	0	35	2026-01-07
ADS10107	44.81	15.29	90.67	-1.04	0	35	2026-01-07
ADS10108	44.6	15.0	89.6	-0.4	0	35	2026-01-07
ADS10109	45.35	14.92	90.09	0.6	0	35	2026-01-07
ADS10110	44.99	15.01	90.01	-0.03	0	35	2026-01-07
ADS10111	45.28	15.0	90.29	0.27	0	35	2026-01-07
ADS10112	45.12	14.79	89.5	0.74	0	35	2026-01-07
ADS10113	45.01	15.36	91.08	-1.06	0	35	2026-01-07
ADS10114	45.01	15.23	90.7	-0.68	0	35	2026-01-07
ADS10115	45.2	15.13	90.59	-0.19	0	35	2026-01-07
ADS10116	45.02	15.02	90.07	-0.02	0	35	2026-01-07
ADS10117	44.81	15.16	90.28	-0.66	0	35	2026-01-07
ADS10118	44.93	15.2	90.53	-0.68	0	35	2026-01-07
ADS10119	44.95	15.22	90.61	-0.7	0	35	2026-01-07
BRM10101	44.82	15.15	90.29	-0.64	0	35	2026-01-07
BRM10102	44.81	15.0	89.8	-0.19	0	35	2026-01-07
BRM10103	45.47	15.0	90.46	0.47	0	35	2026-01-07
BRM10104	44.87	15.05	90.02	-0.27	0	35	2026-01-07
BRM10105	44.5	14.8	88.91	0.09	0	35	2026-01-07
BRM10106	44.21	14.95	89.06	-0.64	0	35	2026-01-07
BRM10107	45.32	14.95	90.16	0.48	0	35	2026-01-07
BRM10108	44.85	15.35	90.9	-1.19	0	35	2026-01-07
BRM10109	44.54	14.89	89.21	-0.14	0	35	2026-01-07
BRM10110	45.02	14.9	89.71	0.33	0	35	2026-01-07
BRM10111	44.25	15.12	89.59	-1.1	0	35	2026-01-07
BRM10112	45.13	15.0	90.14	0.13	0	35	2026-01-07
BRM10113	45.16	15.26	90.95	-0.62	0	35	2026-01-07
BRM10114	44.78	15.32	90.74	-1.19	0	35	2026-01-07
BRM10115	45.34	15.13	90.74	-0.05	0	35	2026-01-07
BRM10116	44.18	15.06	89.35	-0.99	0	35	2026-01-07
BRM10117	45.27	14.78	89.6	0.93	0	35	2026-01-07
BRM10118	44.67	15.32	90.63	-1.28	0	35	2026-01-07
BRM10119	45.06	14.79	89.43	0.7	0	35	2026-01-07
CHM10101	44.76	15.3	90.65	-1.13	0	35	2026-01-07
CHM10102	44.92	14.8	89.31	0.52	0	35	2026-01-07
CHM10103	44.25	14.97	89.16	-0.66	0	35	2026-01-07
CHM10104	44.58	15.43	90.88	-1.71	0	35	2026-01-07
CHM10105	45.23	15.45	91.59	-1.13	0	35	2026-01-07
CHM10106	44.87	14.95	89.74	0.01	0	35	2026-01-07
CHM10107	45.08	15.16	90.54	-0.39	0	35	2026-01-07
CHM10108	44.85	15.06	90.03	-0.32	0	35	2026-01-07
CHM10109	45.32	15.01	90.35	0.29	0	35	2026-01-07
CHM10110	44.6	15.23	90.29	-1.09	0	35	2026-01-07
CPG10101	45.3	15.18	90.85	-0.25	0	35	2026-01-07
CPG10102	44.95	14.65	88.9	1.0	0	35	2026-01-07
CPG10103	45.18	14.94	89.99	0.37	0	35	2026-01-07
CPG10104	44.85	15.13	90.24	-0.54	0	35	2026-01-07
CPG10105	45.09	15.14	90.5	-0.33	0	35	2026-01-07
CPG10106	45.1	14.93	89.89	0.31	0	35	2026-01-07
CPG10107	45.05	14.9	89.74	0.35	0	35	2026-01-07
CPG10108	44.83	14.83	89.31	0.34	0	35	2026-01-07
CPG10109	44.84	14.99	89.81	-0.13	0	35	2026-01-07
CPG10110	45.14	15.0	90.15	0.13	0	35	2026-01-07
CPG10111	44.56	14.7	88.66	0.46	0	35	2026-01-07
CPG10112	44.93	15.05	90.07	-0.21	0	35	2026-01-07
CPG10113	44.8	14.95	89.63	-0.04	0	35	2026-01-07
CPG10114	44.92	15.42	91.18	-1.33	0	35	2026-01-07
CPG10115	45.25	14.6	89.06	1.45	0	35	2026-01-07
CPG10116	45.24	15.3	91.15	-0.68	0	35	2026-01-07
CPG10117	45.35	14.72	89.52	1.17	0	35	2026-01-07
CPG10118	45.05	14.86	89.62	0.47	0	35	2026-01-07
CPG10119	44.98	15.0	89.98	-0.01	0	35	2026-01-07
LYP10101	45.06	15.11	90.4	-0.28	0	35	2026-01-07
LYP10102	44.72	14.98	89.66	-0.22	0	35	2026-01-07
LYP10103	45.19	15.27	90.99	-0.61	0	35	2026-01-07
LYP10104	44.99	15.5	91.49	-1.51	0	35	2026-01-07
LYP10105	45.04	15.36	91.12	-1.03	0	35	2026-01-07
LYP10106	45.08	14.92	89.84	0.32	0	35	2026-01-07
LYP10107	44.69	15.34	90.73	-1.34	0	35	2026-01-07
LYP10108	45.14	14.99	90.1	0.18	0	35	2026-01-07
LYP10109	45.22	14.93	90.02	0.42	0	35	2026-01-07
LYP10110	45.18	15.1	90.47	-0.11	0	35	2026-01-07
LYP10111	45.23	15.52	91.79	-1.33	0	35	2026-01-07

LYP10112	45.19	15.0	90.2	0.18	0	35	2026-01-07
LYP10113	44.74	14.98	89.68	-0.19	0	35	2026-01-07
LYP10114	44.31	15.51	90.83	-2.21	0	35	2026-01-07
LYP10115	45.48	14.87	90.1	0.86	0	35	2026-01-07
LYP10116	44.85	14.88	89.5	0.2	0	35	2026-01-07
LYP10117	45.02	14.78	89.37	0.66	0	35	2026-01-07
LYP10118	45.14	15.2	90.74	-0.47	0	35	2026-01-07
LYP10119	44.92	14.89	89.59	0.26	0	35	2026-01-07
MKC10101	45.05	14.89	89.71	0.39	0	35	2026-01-07
MKC10102	45.27	15.25	91.03	-0.48	0	35	2026-01-07
MKC10103	44.89	14.91	89.63	0.15	0	35	2026-01-07
MKC10104	45.08	15.09	90.35	-0.18	0	35	2026-01-07
MKC10105	45.12	15.07	90.32	-0.09	0	35	2026-01-07
MKC10106	45.43	15.13	90.83	0.03	0	35	2026-01-07
MKC10107	45.12	14.98	90.06	0.18	0	35	2026-01-07
MKC10108	44.74	15.12	90.09	-0.61	0	35	2026-01-07
MKC10109	45.59	15.23	91.29	-0.11	0	35	2026-01-07
MKC10110	44.99	14.85	89.55	0.43	0	35	2026-01-07
MKC10111	45.09	15.19	90.67	-0.49	0	35	2026-01-07
MKC10112	45.02	14.8	89.43	0.61	0	35	2026-01-07
MKC10113	45.08	15.44	91.4	-1.24	0	35	2026-01-07
MKC10114	44.99	14.91	89.73	0.25	0	35	2026-01-07
MKC10115	44.95	14.9	89.65	0.25	0	35	2026-01-07
MKC10116	45.19	14.86	89.76	0.62	0	35	2026-01-07
MKC10117	44.89	14.93	89.67	0.11	0	35	2026-01-07
MKC10118	44.38	14.6	88.18	0.57	0	35	2026-01-07
MKC10119	45.04	15.13	90.43	-0.35	0	35	2026-01-07
MKR10101	44.95	14.72	89.11	0.8	0	35	2026-01-07
MKR10102	44.93	15.09	90.2	-0.34	0	35	2026-01-07
MKR10103	45.2	14.83	89.7	0.71	0	35	2026-01-07
MKR10104	45.2	15.01	90.23	0.16	0	35	2026-01-07
MKR10105	44.97	15.24	90.68	-0.74	0	35	2026-01-07
MKR10106	45.41	14.95	90.26	0.55	0	35	2026-01-07
MKR10107	44.73	14.99	89.71	-0.25	0	35	2026-01-07
MKR10108	45.01	15.3	90.93	-0.9	0	35	2026-01-07
MKR10109	44.94	14.78	89.29	0.59	0	35	2026-01-07
MKR10110	45.09	15.23	90.79	-0.61	0	35	2026-01-07
MKR10111	44.98	14.87	89.58	0.38	0	35	2026-01-07
MKR10112	45.1	15.15	90.54	-0.34	0	35	2026-01-07
MKR10113	45.16	15.09	90.43	-0.12	0	35	2026-01-07
MKR10114	44.55	14.82	89.0	0.1	0	35	2026-01-07
MKR10115	45.31	15.13	90.69	-0.07	0	35	2026-01-07
MKR10116	44.56	14.9	89.26	-0.15	0	35	2026-01-07
MKR10117	44.66	15.08	89.9	-0.58	0	35	2026-01-07
MKR10118	44.96	14.89	89.63	0.29	0	35	2026-01-07
MKR10119	44.78	14.93	89.58	-0.02	0	35	2026-01-07
MSP10101	44.89	14.96	89.77	0.01	0	35	2026-01-07
MSP10102	44.66	15.58	91.41	-2.09	0	35	2026-01-07
MSP10103	44.66	15.04	89.77	-0.46	0	35	2026-01-07
MSP10104	45.29	14.82	89.75	0.83	0	35	2026-01-07
MSP10105	44.95	15.36	91.02	-1.11	0	35	2026-01-07
MSP10106	45.24	15.02	90.3	0.18	0	35	2026-01-07
MSP10107	45.35	14.95	90.21	0.49	0	35	2026-01-07
MSP10108	45.0	15.64	91.9	-1.91	0	35	2026-01-07
MSP10109	45.11	15.09	90.38	-0.16	0	35	2026-01-07
MSP10110	45.38	14.6	89.16	1.59	0	35	2026-01-07
MSR10101	45.09	15.34	91.1	-0.92	0	35	2026-01-07
MSR10102	44.58	15.49	91.06	-1.9	0	35	2026-01-07
MSR10103	44.9	15.06	90.09	-0.29	0	35	2026-01-07
MSR10104	44.96	15.02	90.03	-0.11	0	35	2026-01-07
MSR10105	45.18	15.21	90.81	-0.46	0	35	2026-01-07
MSR10106	44.94	15.39	91.12	-1.24	0	35	2026-01-07
MSR10107	44.9	15.09	90.16	-0.37	0	35	2026-01-07
MSR10108	44.86	15.32	90.82	-1.1	0	35	2026-01-07
MSR10109	45.09	14.53	88.67	1.5	0	35	2026-01-07
MSR10110	45.09	15.6	91.89	-1.71	0	35	2026-01-07
MSR10111	45.02	15.04	90.15	-0.1	0	35	2026-01-07
MSR10112	45.16	14.85	89.71	0.61	0	35	2026-01-07
MSR10113	45.21	14.76	89.49	0.93	0	35	2026-01-07
MSR10114	44.72	15.08	89.97	-0.52	0	35	2026-01-07
MSR10115	45.0	14.82	89.47	0.52	0	35	2026-01-07
MSR10116	44.65	15.42	90.92	-1.62	0	35	2026-01-07
MSR10117	44.98	15.26	90.76	-0.8	0	35	2026-01-07
MSR10118	45.21	15.27	91.01	-0.58	0	35	2026-01-07
MSR10119	44.96	15.23	90.66	-0.74	0	35	2026-01-07

SDV10101	44.87	15.17	90.38	-0.63	0	35	2026-01-07
SDV10102	45.04	15.02	90.1	-0.02	0	35	2026-01-07
SDV10103	44.81	15.17	90.31	-0.69	0	35	2026-01-07
SDV10104	45.09	15.32	91.05	-0.86	0	35	2026-01-07
SDV10105	44.86	14.81	89.3	0.41	0	35	2026-01-07
SDV10106	45.51	15.04	90.61	0.4	0	35	2026-01-07
SDV10107	44.63	15.1	89.93	-0.67	0	35	2026-01-07
SDV10108	45.13	15.3	91.02	-0.77	0	35	2026-01-07
SDV10109	45.11	15.2	90.73	-0.5	0	35	2026-01-07
SDV10110	44.81	14.94	89.64	-0.02	0	35	2026-01-07
SDV10111	45.12	15.34	91.15	-0.9	0	35	2026-01-07
SDV10112	45.09	15.11	90.42	-0.24	0	35	2026-01-07
SDV10113	44.78	15.23	90.47	-0.92	0	35	2026-01-07
SDV10114	45.12	15.1	90.43	-0.2	0	35	2026-01-07
SDV10115	45.18	14.85	89.74	0.63	0	35	2026-01-07
SDV10116	44.76	15.21	90.39	-0.87	0	35	2026-01-07
SDV10117	45.21	15.22	90.86	-0.43	0	35	2026-01-07
SDV10118	45.1	14.97	90.01	0.19	0	35	2026-01-07
SDV10119	45.17	15.05	90.31	0.03	0	35	2026-01-07
SF010101	44.8	14.86	89.39	0.22	0	35	2026-01-07
SF010102	44.8	14.99	89.77	-0.17	0	35	2026-01-07
SF010103	44.92	14.9	89.61	0.22	0	35	2026-01-07
SF010104	44.99	14.89	89.66	0.32	0	35	2026-01-07
SF010105	44.79	15.09	90.07	-0.48	0	35	2026-01-07
SF010106	44.82	15.01	89.86	-0.21	0	35	2026-01-07
SF010107	44.44	14.83	88.94	-0.06	0	35	2026-01-07
SF010108	45.12	15.96	92.98	-2.75	0	35	2026-01-07
SF010109	45.22	15.25	90.97	-0.54	0	35	2026-01-07
SF010110	44.89	15.14	90.32	-0.54	0	35	2026-01-07
SSP10101	44.62	15.24	90.35	-1.11	0	35	2026-01-07
SSP10102	45.19	14.93	89.97	0.42	0	35	2026-01-07
SSP10103	44.9	14.97	89.82	-0.03	0	35	2026-01-07
SSP10104	44.81	14.83	89.3	0.31	0	35	2026-01-07
SSP10105	45.08	15.14	90.5	-0.34	0	35	2026-01-07
SSP10106	45.16	14.75	89.4	0.91	0	35	2026-01-07
SSP10107	45.48	15.34	91.5	-0.55	0	35	2026-01-07
SSP10108	44.67	15.0	89.66	-0.33	0	35	2026-01-07
SSP10109	45.82	15.18	91.35	0.29	0	35	2026-01-07
SSP10110	44.96	14.99	89.92	-0.0	0	35	2026-01-07

=== MKTS_Module_model_summary.txt ===

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 8.933069603449873e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: MKC10113, MKC10106, MKR10000. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CPG10107, MSR10107, CPG10102. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: CPG10107, MKR10114, SDV10105. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- MKC10113: Cost Per Click (CPC)
- MKC10106: Open Rate
- MKR10000: Market Research

Scenario_9 (Avg Risk=0.804):

- CPG10107: Return on Marketing Investment (ROMI)
- MSR10107: Customer Support Response Time
- CPG10102: Impressions

Scenario_8 (Avg Risk=0.639):

- CPG10107: Return on Marketing Investment (ROMI)
- MKR10114: Market Potential
- SDV10105: Customer Acquisition Cost (CAC)

Baseline Predictor (Mean of MKTS_001):

MSE: 1.223451667988889

R-squared: 0.0

RMSE: 1.1060974947936955

MAPE: 1.9635913214388674%

PyTorch Model Evaluation:

MSE: 0.0006558244433992936

R-squared: 0.9994639555770296

RMSE: 0.02560906955356429

MAPE: 0.04542812086958068%

Model Evaluation Metrics:

MSE: 0.0006558244433992936

RMSE: 0.02560906955356429

R²: 0.9994639555770296

MAPE: 0.04542812086958068%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of MKTS Module Risk Analysis Summary-----